UP-PCS PRELIMS 2026 Â· HIGH PROBABILITY QUESTION BANK

Master Expected Questions — Part 5

High-Yield Questions 301 to 375 (Random Subject Distribution with UPPSC Traps & Ratta Points)

QUESTION RANGE
Questions 301 – 375

PATTERN & STYLE
UPPSC 2024â€"2025 Standard (Statements, A&R, Match Lists)

STUDY FORMAT
Question â†' Options â†' Answer â†' In-Depth Ratta & Trap Notes

TARGET DATE
December 6 Prelims Examination

UPPCS Prelims 2026 Expected Questions — Part 5 (Q301 to Q375)

Subject Distribution: Randomized Full-Spectrum GS (History, Geography, Polity, Science, Environment, Economy, UP Special, Current Affairs)
Exam Standard: Multi-statement, Assertion-Reason, Match Lists & Direct Factual Traps as per UPPSC 2024–2025 Trends

Question 301

Subject: Modern Indian History | **Topic:** Dandi March & Civil Disobedience Movement (1930)

Consider the following statements regarding the historic Dandi March: 1. Mahatma Gandhi began the march from Sabarmati Ashram on March 12, 1930, accompanied by 78 chosen ashram followers. 2. The march covered a distance of approximately 241 miles (385 km) over 24 days, reaching Dandi on April 5, 1930. 3. On the morning of April 6, 1930, Gandhi broke the Salt Law by picking up a lump of natural salt, formally inaugurating the Civil Disobedience Movement.

Which of the statements given above are correct?

Options: A. 1 and 2 only
B. 2 and 3 only
C. 1 and 3 only
D. 1, 2 and 3

Answer: D

Explanation: - **Core Fact & Solution:** - All three statements are correct: - **March Launch:** Started March 12, 1930, from Sabarmati Ashram with 78 volunteers; arrived at coastal Dandi (Navsari district, Gujarat) on April 5; broke the salt monopoly on **April 6, 1930**. - **American Journalist:** Webb Miller of United Press famously documented the heroic non-violent protest at Dharasana Salt Works (where Sarojini Naidu, Imam Saheb, and Manilal Gandhi led satyagrahis against brutal police lathi charges). - **Regional Parallels:** - Tamil Nadu: C. Rajagopalachari marched from Trichinopoly to Vedaranyam (*Vedaranyam Salt March*). - Malabar (Kerala): K. Kelappan ("Kerala Gandhi") marched from Calicut to Payyanur. - Assam: Satyagrahis marched from Sylhet to Noakhali. - NWFP: Khan Abdul Ghaffar Khan ("Frontier Gandhi") led the *Khudai Khidmatgars* (Red Shirts). - **आयोग का Trap / Ratta Point:** Sarojini Naidu accompanied Gandhi at Dandi and famously hailed him: "*Hail, Deliverer!*"

Question 302

Subject: Indian Polity | **Topic:** Pardoning Powers: President (Article 72) vs Governor (Article 161)

Consider the following statements comparing the pardoning powers of the President and the Governor: 1. The President can grant pardon, reprieves, respites, or remissions of punishment in all cases where the sentence is by a Court Martial (Military Court), whereas the Governor has no such power. 2. The President is the exclusive constitutional authority empowered to grant pardon in cases of death sentence; the Governor CANNOT pardon a death sentence. 3. The Governor can suspend, remit, or commute a sentence of death under Article 161.

Which of the statements given above are correct?

Options: A. 1 and 2 only
B. 2 and 3 only
C. 1 and 3 only
D. 1, 2 and 3

Answer: D

Explanation: - **Core Fact & Solution:** - All three statements are correct: - **Court Martial:** President alone possesses the power to pardon court-martial sentences (Article 72(1)(a)). Governor has zero power regarding military courts. - **Death Sentence:** Under Article 72, the **President alone can PARDON a death sentence** (completely wiping out guilt and sentence). - **Governor's Power regarding Death Sentence:** Under Article 161, the Governor CANNOT pardon a death sentence, BUT the Governor **CAN suspend, remit, or commute** a death sentence under state law! - **आयोग का Trap / Ratta Point:** Landmark ruling *Maru Ram (1980)* and *Kehar Singh (1989)*: The pardoning power under Articles 72 and 161 is exercised by the President/Governor on the **binding advice of the Council of Ministers**, not at personal discretion!

Question 303

Subject: Geography of India | **Topic:** Brahmaputra River System & Tributaries

Match List-I (Tributary of Brahmaputra) with List-II (Bank Orientation & Origin / Feature) and select the correct answer:

List-I (Tributary) A. Subansiri
B. Manas
C. Lohit
D. Teesta

List-II (Orientation & Feature) 1. Left-bank tributary; originates in eastern Tibet; passes through Mishmi Hills; known as "River of Blood"
2. Right-bank tributary; originates in Kangchenjunga range/Pahunri glacier (Sikkim); joins Brahmaputra in Bangladesh
3. Right-bank tributary; antecedent river originating in Tibet; largest tributary of Brahmaputra ("Gold River")
4. Right-bank transboundary river originating in Bhutan; flows through Manas National Park

Options: A. A-3, B-4, C-1, D-2
B. A-1, B-4, C-3, D-2
C. A-3, B-2, C-1, D-4
D. A-4, B-3, C-1, D-2

Answer: A

Explanation: - **Core Fact & Solution:** - **Subansiri ("Gold River"):** Largest tributary of Brahmaputra; antecedent river from Tibet; joins from the **Right bank**. - **Manas:** Originates in Bhutan; joins Brahmaputra from the **Right bank** at Jogighopa. - **Lohit:** Originates in Tibet (Zayal Chu); flows through Arunachal; joins Dibang and Dihang at Sadiya from the **Left bank**; carries red laterite silt ("River of Blood"); Dhola-Sadiya Bridge (Bhupen Hazarika Setu, 9.15 km) is built on it. - **Teesta:** Originates from Tso Lhamo lake / Pahunri glacier in Sikkim; historically flowed into Ganga, but switched eastward into Brahmaputra (Jamuna) after the 1787 mega-floods. - **आयोग का Trap / Ratta Point:** Left-bank tributaries: **Dibang, Lohit, Dhansiri, Burhi Dihing, Kalang**. Right-bank tributaries: **Subansiri, Kameng (Jia Bhoreli), Manas, Sankosh, Teesta**!

Question 304

Subject: General Science (Biology & Physics) | **Topic:** Pathologies of the Human Eye

Match List-I (Eye Disorder) with List-II (Pathological Cause / Nature) and select the correct answer:

List-I (Disorder) A. Presbyopia
B. Astigmatism
C. Cataract
D. Glaucoma

List-II (Pathology) 1. Progressive clouding/opacity of the crystalline eye lens, blocking light from reaching the retina
2. Loss of elasticity of the eye lens and gradual weakening of ciliary muscles with aging
3. Increased intraocular fluid pressure (aqueous humor) causing irreversible damage to the optic nerve
4. Aspherical / uneven curvature of the cornea or lens, causing light to focus at multiple points

Options: A. A-2, B-4, C-1, D-3
B. A-4, B-2, C-1, D-3
C. A-2, B-1, C-4, D-3
D. A-3, B-4, C-1, D-2

Answer: A

Explanation: - **Core Fact & Solution:** - **Presbyopia:** Aging defect (usually after age 40); ciliary muscles lose contractility; near-point recedes; corrected with **Bifocal lenses**. - **Astigmatism:** Cornea is curved like a rugby ball rather than a spherical football; horizontal and vertical rays focus at different planes; corrected with **Cylindrical lenses**. - **Cataract (Motiyabind):** Age-related denaturing of lens proteins (*crystallins*); lens becomes cloudy/opaque; treated surgically via phacoemulsification and Intraocular Lens (IOL) implantation. - **Glaucoma (Kala Motia):** Obstruction in the drainage canal (Canal of Schlemm) leads to elevated Intraocular Pressure (IOP > 21 mm Hg), permanently damaging optic nerve axons. - **आयोग का Trap / Ratta Point:** Myopia uses Concave lens; Hypermetropia uses Convex lens; Presbyopia uses Bifocal lens; Astigmatism uses Cylindrical lens!

Question 305

Subject: Uttar Pradesh Special | **Topic:** Literacy Demographics of UP (Census 2011)

Consider the following statements regarding literacy rates in Uttar Pradesh: 1. As per Census 2011, the overall literacy rate of Uttar Pradesh is 67.72%. 2. The male literacy rate in Uttar Pradesh is 77.28%, while the female literacy rate is 57.18%. 3. The male-female literacy gap in Uttar Pradesh stands at approximately 20.10 percentage points.

Which of the statements given above are correct?

Options: A. 1 and 2 only
B. 2 and 3 only
C. 1 and 3 only
D. 1, 2 and 3

Answer: D

Explanation: - **Core Fact & Solution:** - All three statements are correct: - **Overall UP Literacy: 67.72%** (Male: 77.28%, Female: 57.18%). - **National Average Comparison:** India's literacy in 2011 was 73.0% (Male: 80.9%, Female: 64.6%). UP ranks lower than national average on all three counts. - **Gender Gap:** The gap between male and female literacy in UP is **20.10%** (Maharajganj has the widest gender gap in literacy). - **Highest Literacy:** Gautam Buddha Nagar (80.12%); Lowest: Shravasti (46.74%). - **आयोग का Trap / Ratta Point:** Mnemonic for UP Literacy: **Overall ~67%, Male (+10%) ~77%, Female (-10%) ~57%**!

Question 306

Subject: Environment & Ecology | **Topic:** Kyoto Protocol (1997) & Market Mechanisms

Consider the following statements regarding the Kyoto Protocol: 1. The Kyoto Protocol was adopted in 1997 and entered into force in February 2005, committing Annex I industrialized nations to legally binding greenhouse gas reduction targets. 2. It established three flexible market-based mechanisms: Clean Development Mechanism (CDM), Joint Implementation (JI), and International Emissions Trading (IET). 3. Under the Clean Development Mechanism (CDM), an Annex I developed country can earn certified emission reduction (CER) credits by investing in emission-reduction projects in non-Annex I developing nations like India.

Which of the statements given above are correct?

Options: A. 1 and 2 only
B. 2 and 3 only
C. 1 and 3 only
D. 1, 2 and 3

Answer: D

Explanation: - **Core Fact & Solution:** - All three statements are correct: - **Adoption & Status:** Adopted at COP3 in Kyoto, Japan (1997); came into force on 16 February 2005 after Russian ratification; operationalized the UNFCCC under the principle of "Common But Differentiated Responsibilities" (CBDR). - **Target GHGs (6 gases):** Carbon Dioxide (CO_2), Methane (CH_4), Nitrous Oxide (N_2O), Hydrofluorocarbons (HFCs), Perfluorocarbons (PFCs), Sulphur Hexafluoride (SF_6). (Nitrogen Trifluoride NF_3 added in Doha Amendment). - **3 Market Mechanisms:** - **CDM (Article 12):** Projects in developing countries earning Certified Emission Reductions (CERs). - **Joint Implementation (Article 6):** Projects between Annex I developed countries. - **International Emissions Trading (Article 17):** Buying/selling excess assigned amount units (AAUs) ("Carbon Credits"). - **आयोग का Trap / Ratta Point:** Developing countries (like India and China) had **NO legally binding emission reduction commitments** under the Kyoto Protocol!

Question 307

Subject: Indian Economy | **Topic:** Taxation Principles & Gini Coefficient

Match List-I (Tax Structure) with List-II (Definition / Impact on Income Inequality) and select the correct answer:

List-I (Tax Structure) A. Progressive Tax
B. Regressive Tax
C. Proportional Tax
D. Degressive Tax

List-II (Definition & Impact) 1. Tax rate remains constant regardless of the taxpayer's income level
2. Tax rate increases as the income increases, reducing the post-tax Gini coefficient (e.g. Personal Income Tax)
3. Tax rate increases up to a certain maximum income limit, after which it remains flat / uniform
4. Effective tax burden falls more heavily on lower-income earners than on the wealthy (e.g. Indirect taxes on essential goods)

Options: A. A-2, B-4, C-1, D-3
B. A-1, B-2, C-4, D-3
C. A-2, B-1, C-4, D-3
D. A-3, B-4, C-1, D-2

Answer: A

Explanation: - **Core Fact & Solution:** - **Progressive Tax:** Marginal tax rate increases as taxable base increases; redistributive; narrows the rich-poor gap (reduces Gini coefficient). - **Regressive Tax:** Poor people spend a higher proportion of their income on consumed goods; thus indirect consumption taxes (like GST or VAT) place a disproportionately higher percentage burden on lower-income earners. - **Proportional Tax (Flat Tax):** Rate remains constant across all income brackets. - **Degressive Tax:** Blend of progressive and proportional; rate rises progressively up to a threshold income, and thereafter remains fixed at that upper rate. - **आयोग का Trap / Ratta Point:** Direct taxes (income/corporate tax) are generally PROGRESSIVE; Indirect taxes (excise/GST) are inherently REGRESSIVE!

Question 308

Subject: Current Affairs | **Topic:** Global Liveability Index 2024

Consider the following statements regarding the Global Liveability Index 2024 published by the Economist Intelligence Unit (EIU): 1. Vienna (Austria) retained its position as the world's most liveable city for the third consecutive year. 2. Damascus (Syria) was ranked as the least liveable city in the world. 3. The index evaluates 173 cities across five broad categories: Stability, Healthcare, Culture & Environment, Education, and Infrastructure.

Which of the statements given above are correct?

Options: A. 1 and 2 only
B. 2 and 3 only
C. 1 and 3 only
D. 1, 2 and 3

Answer: D

Explanation: - **Core Fact & Solution:** - All three statements are correct: - **1st Rank: Vienna, Austria** scored near-perfect 98.4 (perfect 100 in stability, healthcare, education, infrastructure), followed by Copenhagen (Denmark), Zurich (Switzerland), and Melbourne (Australia). - **Bottom Rank: Damascus, Syria** ranked 173rd (score 30.7), followed by Tripoli (Libya) and Algiers (Algeria). - **Indian Cities:** New Delhi and Mumbai tied at **140th rank**; Bengaluru ranked 146th. - **आयोग का Trap / Ratta Point:** Published by the **Economist Intelligence Unit (EIU)**, NOT Mercer or UN-Habitat! (Mercer publishes the Quality of Living City Ranking).

Question 309

Subject: Ancient Indian History | **Topic:** Mauryan Rock-Cut Architecture (Barabar Caves)

Consider the following statements regarding the Barabar and Nagarjuni rock-cut caves: 1. The Barabar caves located in Jehanabad district of Bihar are the oldest surviving rock-cut cave monuments in India, dating to the Mauryan period. 2. Emperor Ashoka dedicated the caves at Barabar (such as Sudama Cave, Karna Chaupar, and Lomas Rishi) for the use of the monks of the Ajivika sect. 3. Ashoka's grandson Dasharatha dedicated the adjacent Nagarjuni caves (such as Gopika cave) to the Ajivikas.

Which of the statements given above are correct?

Options: A. 1 and 2 only
B. 2 and 3 only
C. 1 and 3 only
D. 1, 2 and 3

Answer: D

Explanation: - **Core Fact & Solution:** - All three statements are correct: - **Barabar Caves (3rd Century BCE):** Monolithic caves carved out of solid granite rock; famous for mirror-like interior glass polish (*Mauryan Polish*). - **Ajivika Dedication:** Inscriptions inside Sudama and Karna Chaupar explicitly state King Piyadasi (Ashoka) dedicated them to the **Ajivika sect** (founded by Makkhali Gosala, propounding *Niyati* / fatalism). - **Lomas Rishi Cave:** Famous for its carved horseshoe-shaped chaitya entrance facade depicting a frieze of elephants bowing before stupas. - **Nagarjuni Caves:** Inscriptions verify they were excavated by Ashoka's grandson **Dasharatha Maurya** for Ajivika ascetics. - **आयोग का Trap / Ratta Point:** Even though Ashoka was a devoted Buddhist, he patronized the AJIVIKA sect by carving caves for them at Barabar, demonstrating genuine religious tolerance (as preached in Major Rock Edict XII)!

Question 310

Subject: World Geography | **Topic:** Local Cold Winds of the World

Match List-I (Local Cold Wind) with List-II (Region / Country) and select the correct answer:

List-I (Cold Wind) A. Bora
B. Mistral

C. Pampero
D. Blizzard

List-II (Region) 1. Rhone Valley (France) and Gulf of Lion
2. Adriatic Coast (Croatia and Italy)
3. North American tundra and prairies (USA / Canada)
4. Pampas of Argentina and Uruguay

Options: A. A-2, B-1, C-4, D-3
B. A-1, B-2, C-4, D-3
C. A-2, B-4, C-1, D-3
D. A-3, B-1, C-4, D-2

Answer: A

Explanation: - **Core Fact & Solution:** - **Bora:** Fierce, gusty, cold, dry northeasterly katabatic wind blowing from the Dinaric Alps down to the **Adriatic Sea coast**. - **Mistral:** Cold, dry northwesterly wind that channels down the **Rhone Valley in southern France** towards the Mediterranean during winter/spring. - **Pampero:** Violent, cold polar wind from the south that sweeps across the **Pampas of Argentina and Uruguay**. - **Blizzard:** Severe cold, snow-laden wind associated with intense polar high pressure systems across **Canada and northern USA**. - **आयोग का Trap / Ratta Point:** *Buran* is a cold blizzard wind blowing over Siberia/Russia; *Khamsin* is a hot, dusty wind in Egypt; *Loo* is a hot dry summer wind in northern India!

Question 311

Subject: Indian Polity | **Topic:** Money Bills (Article 110)

Consider the following statements regarding a Money Bill in the Indian Parliament: 1. Under Article 110, a Bill is deemed to be a Money Bill if it contains ONLY provisions dealing with taxation, borrowings, custody of the Consolidated Fund of India, or matters incidental thereto. 2. A Money Bill can be introduced ONLY in the Lok Sabha, and ONLY on the prior recommendation of the President of India. 3. The Rajya Sabha has no power to amend or reject a Money Bill; it can only make recommendations, which the Lok Sabha may accept or reject, within a mandatory period of 14 days. 4. If a question arises whether a Bill is a Money Bill or not, the decision of the Speaker of the Lok Sabha thereon is final.

Which of the statements given above are correct?

Options: A. 1, 2 and 3 only
B. 2, 3 and 4 only
C. 1, 3 and 4 only
D. 1, 2, 3 and 4

Answer: D

Explanation: - **Core Fact & Solution:** - All four statements are correct: - **Definition (Article 110(1)):** Enlists 7 specific fiscal matters (imposition, abolition, remission, alteration, regulation of any tax; government borrowings; custody of Consolidated or Contingency Fund). - **Introduction (Article 109 & 117(1)):** Originates ONLY in Lok Sabha with President's recommendation. Cannot be introduced in Rajya Sabha. - **Rajya Sabha Role:** RS cannot amend or reject; has strictly **14 days** to return the bill. If not returned in 14 days, it is deemed passed by both Houses. - **Speaker's Endorsement:** Speaker signs a certificate when transmitting it to RS and President. - **Joint Sitting:** There is **NO provision for a Joint Sitting (Article 108)** for a Money Bill! - **President's Assent (Article 111):** President may give assent or withhold assent, but **CANNOT return a Money Bill** for reconsideration. - **आयोग का Trap / Ratta Point:** Joint sitting is available ONLY for Ordinary Bills and Financial Bills (Category A/B), NEVER for Money Bills or Constitution Amendment Bills!

Question 312

Subject: Medieval Indian History | **Topic:** Major Battles of the Sultanate & Early Mughals

Match List-I (Battle) with List-II (Year & Belligerents) and select the correct answer:

List-I (Battle) A. Battle of Chandawar
B. First Battle of Panipat
C. Battle of Chanderi
D. Battle of Ghaghra

List-II (Year & Belligerents) 1. 1526; Babur defeated Ibrahim Lodi, ending Delhi Sultanate
2. 1194; Muhammad Ghori defeated Jaichand, King of Kannauj (Gahadavala dynasty)
3. 1529; Babur defeated the joint Afghan forces under Mahmud Lodi and Nusrat Shah of Bengal
4. 1528; Babur defeated Medini Rai, ruler of Malwa

Options: A. A-2, B-1, C-4, D-3
B. A-1, B-2, C-4, D-3
C. A-2, B-4, C-1, D-3
D. A-3, B-1, C-4, D-2

Answer: A

Explanation: - **Core Fact & Solution:** - **Battle of Chandawar (1194):** Fought on Yamuna river near Firozabad; Muhammad Ghori defeated **Raja Jaichand of Kannauj**; established Turkish rule across the Gangetic plain. - **First Battle of Panipat (21 April 1526):** Babur introduced firearms, gunpowder artillery, and *Tulghuma* flanking tactics to defeat Ibrahim Lodi. - **Battle of Chanderi (1528):** Babur defeated Rajput chieftain **Medini Rai** of Chanderi. - **Battle of Ghaghra (May 1529):** Babur defeated allied Afghan forces of Bihar and Nusrat Shah of Bengal; was Babur's last battle. - **आयोग का Trap / Ratta Point:** Babur's 4 consecutive annual victories: **1526 Panipat -> 1527 Khanwa -> 1528 Chanderi -> 1529 Ghaghra**! (Babur died at Agra in December 1530).

Question 313

Subject: General Science (Chemistry) | **Topic:** Hydrated Inorganic Salts (Vitriols)

Match List-I (Common Name) with List-II (Chemical Formula & Metal) and select the correct answer:

List-I (Salt) A. Blue Vitriol
B. Green Vitriol
C. White Vitriol
D. Epsom Salt

List-II (Formula) 1. Ferrous Sulphate Heptahydrate ($FeSO_4 \cdot 7H_2O$)
2. Copper Sulphate Pentahydrate ($CuSO_4 \cdot 5H_2O$)
3. Magnesium Sulphate Heptahydrate ($MgSO_4 \cdot 7H_2O$)
4. Zinc Sulphate Heptahydrate ($ZnSO_4 \cdot 7H_2O$)

Options: A. A-2, B-1, C-4, D-3
B. A-1, B-2, C-4, D-3
C. A-2, B-4, C-1, D-3
D. A-3, B-1, C-4, D-2

Answer: A

Explanation: - **Core Fact & Solution:** - **Blue Vitriol (Neela Thotha):** $CuSO_4 \cdot 5H_2O$ (Copper Sulphate Pentahydrate); fungicide; used in Bordeaux mixture with slaked lime. - **Green Vitriol (Hara Kasis):** $FeSO_4 \cdot 7H_2O$ (Ferrous Sulphate Heptahydrate); used in iron supplements and dyes. - **White Vitriol:** $ZnSO_4 \cdot 7H_2O$ (Zinc Sulphate Heptahydrate); used in eye lotions and skin ointments. - **Epsom Salt:** $MgSO_4 \cdot 7H_2O$ (Magnesium Sulphate Heptahydrate); used as a saline purgative/laxative. - **Oil of Vitriol:** Concentrated Sulphuric Acid (H_2SO_4). - **आयोग का Trap / Ratta Point:** Blue Vitriol = **Copper**; Green Vitriol = **Iron**; White Vitriol = **Zinc**; Epsom Salt = **Magnesium**!

Question 314

Subject: Uttar Pradesh Special | **Topic:** Folk Dances and Cultural Traditions

Consider the following statements regarding traditional folk arts of Uttar Pradesh: 1. 'Charkula' is a spectacular folk dance of the Braj region where women balance a multi-tiered wooden structure with 108 illuminated oil lamps on their heads. 2. 'Kajari' is a classical-folk monsoon song originating from the Mirzapur-Varanasi region. 3. 'Alha' singing in Bundelkhand narrates the chivalrous exploits of the 12th-century Banafar warrior heroes Alha and Udal.

Which of the statements given above are correct?

Options: A. 1 and 2 only
B. 2 and 3 only
C. 1 and 3 only
D. 1, 2 and 3

Answer: D

Explanation: - **Core Fact & Solution:** - All three statements are correct: - **Charkula Dance:** Practiced in Braj (Mathura/Vrindavan) on the third day after Holi (Radha's birthday); dancers veil their faces and carry 108 oil lamps arranged on 4-5 circular wooden tiers (*charkula*). - **Kajari:** Sung during the rainy season (Shravana); portrays the agony of separation from the beloved; two forms: *Dharau* (sung at home by women) and *Akhade ki Kajari* (sung competitively on stage). - **Alha:** Composed originally in Bundeli/Bhojpuri by court poet Jagnik (*Alha-Khand* / *Parmal Raso*); recounts 52 epic battles of generals Alha and Udal. - **आयोग का Trap / Ratta Point:** Dhobia

dance is performed by the washermen community of Eastern UP; Karma dance is performed by the Kharwar and Kol tribes of Sonbhadra!

Question 315

Subject: Environment & Ecology | **Topic:** Food Chains & Trophic Energy Flow

Consider the following statements regarding Grazing Food Chains (GFC) and Detritus Food Chains (DFC): 1. In a Grazing Food Chain, the primary energy source is solar radiation captured by living green plants (producers). 2. In a Detritus Food Chain, the initial energy source is dead organic matter (detritus) decomposed by microorganisms (fungi and bacteria). 3. In terrestrial ecosystems, a much larger fraction of energy flows through the Detritus Food Chain than through the Grazing Food Chain.

Which of the statements given above are correct?

Options: A. 1 and 2 only
B. 2 and 3 only
C. 1 and 3 only
D. 1, 2 and 3

Answer: D

Explanation: - **Core Fact & Solution:** - All three statements are correct: - **GFC:** Living Plants $ ightarrow$ Herbivores $ ightarrow$ Primary Carnivores $ ightarrow$ Top Carnivores. In aquatic ecosystems (oceans/lakes), GFC is the major conduit for energy flow. - **DFC:** Dead Plant/Animal Matter $ ightarrow$ Decomposers (bacteria/fungi) / Detritivores (earthworms, millipedes). Decomposers secrete extracellular digestive enzymes to break down organic matter into inorganic nutrients (*Mineralization*). - **Terrestrial Energy Flow:** In forests and grasslands, only ~10% of plant biomass is consumed while green by herbivores; the remaining **~90% dies and enters the Detritus Food Chain**, making DFC the dominant pathway for energy and nutrient cycling on land. - **आयोग का Trap / Ratta Point:** In Aquatic ecosystems, GFC is dominant; in Terrestrial ecosystems, DFC is dominant!

Question 316

Subject: Indian Economy | **Topic:** Foreign Direct Investment (FDI) Policy

Under the Consolidated FDI Policy of the Government of India, in which of the following sectors is Foreign Direct Investment (FDI) completely **PROHIBITED** under both the automatic and government approval routes? 1. Lottery Business (including government, private, and online lotteries) 2. Gambling and Betting (including casinos) 3. Atomic Energy generation 4. Cigars, cheroots, cigarillos, and cigarettes manufacturing

Select the correct answer from the codes given below:

Options: A. 1, 2 and 3 only
B. 1, 2 and 4 only
C. 2, 3 and 4 only
D. 1, 2, 3 and 4

Answer: D

Explanation: - **Core Fact & Solution:** - Under Government of India regulations, FDI is strictly **PROHIBITED in 9 specific sectors**: 1. Lottery business (government/private/online) 2. Gambling and betting (including casinos) 3. Chit funds 4. Nidhi company 5. Trading in Transferable Development Rights (TDRs) 6. Real Estate business or construction of farm houses (excluding township developments) 7. Manufacturing of cigars, cheroots, cigarillos, and cigarettes (tobacco products) 8. Activities / sectors not open to private sector investment (e.g., **Atomic Energy**) 9. Railway operations (except dedicated railway infrastructure/high-speed lines, which permit 100% FDI). - **आयोग का Trap / Ratta Point:** Defense manufacturing permits up to 74% FDI under the automatic route (and up to 100% via government approval)!

Question 317

Subject: Current Affairs | **Topic:** 69th National Film Awards

At the 69th National Film Awards, who became the first Telugu actor to win the National Film Award for Best Actor?

Options: A. Allu Arjun
B. Ram Charan
C. Jr. NTR
D. Prabhas

Answer: A

Explanation: - **Core Fact & Solution:** - **Allu Arjun** won the National Film Award for **Best Actor** for his titular role in the blockbuster Telugu film *Pushpa: The Rise (Part 1)* directed by Sukumar. - He created history as the **first-ever Telugu actor** to win the Best Actor award in the 69-year history of the National Film Awards. - Best Actress was shared by **Alia Bhatt** (*Gangubai Kathiawadi*) and **Kriti Sanon** (*Mimi*). - Best Feature Film went to *Rocketry: The Nambi Effect* (directed by R. Madhavan). - Best Popular Film Providing Wholesome Entertainment went to *RRR* (directed by S.S. Rajamouli). - **आयोग का Trap / Ratta Point:** Best Music Direction at the 69th National Film Awards was won by **M.M. Keeravani** for *RRR* and **Devi Sri Prasad** for *Pushpa*!

Question 318

Subject: Modern Indian History | **Topic:** Administration of Lord Ripon (1880–1884)

Consider the following statements regarding Viceroy Lord Ripon: 1. He is known as the "Father of Local Self-Government in India" due to his landmark Resolution of May 1882. 2. He repealed Lord Lytton's repressive Vernacular Press Act in 1882. 3. The First Factory Act of 1881, which regulated child labor in mechanized factories, was enacted during his tenure. 4. The Ilbert Bill controversy (1883) erupted during his viceroyalty over granting Indian magistrates parity to try Europeans in criminal cases.

Which of the statements given above are correct?

Options: A. 1, 2 and 3 only
B. 2, 3 and 4 only
C. 1, 3 and 4 only
D. 1, 2, 3 and 4

Answer: D

Explanation: - **Core Fact & Solution:** - All four statements are correct: - Lord Ripon (appointed by Gladstone's Liberal government) was hailed by Florence Nightingale as the "*Saviour of India*": - **Resolution of 1882:** Laid down the framework of sub-divisional, district, and municipal local self-governing boards with elected non-official majorities. - **Repeal of Vernacular Press Act (1882):** Restored press equality between Indian and Anglo-Indian newspapers. - **Factory Act of 1881:** Banned employment of children under 7; capped working hours of children aged 7–12 at 9 hours/day. - **Ilbert Bill (1883):** Drafted by Law Member Sir Courtenay Ilbert; violent racist agitation by European settlers (*White Mutiny*) forced the government to amend it, allowing Europeans to demand trial by a jury with at least 50% Europeans. - **आयोग का Trap / Ratta Point:** First official Decennial Census of India was conducted under Lord Ripon in **1881**! (Lord Mayo had initiated the first non-synchronous census in 1872).

Question 319

Subject: Geography of India | **Topic:** Physiographic Plateaus of India

Match List-I (Plateau Region) with List-II (Distinct Geographical Feature) and select the correct answer:

List-I (Plateau) A. Malwa Plateau
B. Chota Nagpur Plateau
C. Meghalaya (Shillong) Plateau
D. Deccan Trap

List-II (Feature) 1. Separated from Rajmahal Hills by Malda Gap; receives world's highest rainfall
2. Step-like volcanic basaltic lava landscape; black cotton soil (Regur)
3. Dissected sandstone plateau drained by Chambal; famous for badland ravines
4. Known as the "Ruhr of India"; richest mineral belt containing Gondwana coal

Options: A. A-3, B-4, C-1, D-2
B. A-1, B-4, C-3, D-2
C. A-3, B-2, C-1, D-4
D. A-4, B-3, C-1, D-2

Answer: A

Explanation: - **Core Fact & Solution:** - **Malwa Plateau:** Lava plateau bounded by Aravallis to the west and Vindhyas to the south; drained by Chambal, Sindh, Betwa, and Ken; heavily gullied into **Chambal Badlands / Ravines**. - **Chota Nagpur Plateau:** "Mineral Heartland / Ruhr of India"; rich in iron, coal (Jharia, Bokaro), mica, and bauxite; drained by Damodar, Subarnarekha, and Barakar rivers. - **Meghalaya (Shillong) Plateau:** An eastward geological extension of the Peninsular shield (NOT the Himalayas!); severed from Chota Nagpur by the **Garo-Rajmahal Gap (Malda Gap)**; houses Garo, Khasi, and Jaintia hills. - **Deccan Trap:** Cretaceous volcanic fissure basaltic eruptions covering Maharashtra, parts of Gujarat and MP; weathering gives rich black soil. - **आयोग का Trap / Ratta Point:** The Shillong (Meghalaya) plateau is structurally part of the **PENINSULAR DECCAN SHIELD**, not the Himalayan orogen!

Question 320

Subject: Indian Polity | **Topic:** Special Constitutional Provisions for States (Articles 371 to 371J)

Match List-I (Article of the Constitution) with List-II (State Covered under Part XXI) and select the correct answer:

List-I (Article) A. Article 371A
B. Article 371C
C. Article 371F
D. Article 371J

List-II (State) 1. Manipur (special committee of Hill Areas)
2. Nagaland (customary law, ownership of land and resources)
3. Karnataka (Hyderabad-Karnataka region)
4. Sikkim (special provisions inserted by 36th CAA 1975)

Options: A. A-2, B-1, C-4, D-3
B. A-1, B-2, C-4, D-3
C. A-2, B-4, C-1, D-3
D. A-3, B-1, C-4, D-2

Answer: A

Explanation: - **Core Fact & Solution:** - High-yield cheat-sheet of **Special Provisions under Articles 371 to 371J**: - **Article 371:** Maharashtra and Gujarat (Vidarbha, Marathwada, Saurashtra, Kutch development boards). - **Article 371A: Nagaland** (Acts of Parliament cannot override Naga customary laws/religious practices without Assembly approval). - **Article 371B:** Assam (committee of tribal areas). - **Article 371C: Manipur** (committee of members from Hill Areas). - **Article 371D & 371E:** Andhra Pradesh and Telangana (equitable educational/employment opportunities). - **Article 371F: Sikkim** (added by 36th CAA 1975). - **Article 371G:** Mizoram (Mizo customary law and land transfer). - **Article 371H:** Arunachal Pradesh (Governor's special responsibility for law and order). - **Article 371I:** Goa (minimum 30 assembly members). - **Article 371J: Karnataka** (added by 98th CAA 2012 for 6 backward districts of Hyderabad-Karnataka: Bellary, Bidar, Gulbarga, Koppal, Raichur, Yadgir). - **आयोग का Trap / Ratta Point:** Mnemonic: **371A = Nagaland, 371C = Manipur, 371F = Sikkim, 371G = Mizoram, 371J = Karnataka**!

Question 321

Subject: Ancient Indian History | **Topic:** Vedic Literature & Social Organization

Consider the following statements regarding the Vedic literature: 1. The 'Purusha Sukta', which describes the fourfold Varna system for the first time, is located in the Tenth Mandala of the Rigveda. 2. The Gayatri Mantra is addressed to the solar deity Savitr and is composed in the Third Mandala of the Rigveda by sage Vishvamitra. 3. The 'Sabha' and 'Samiti' were described as the two twin daughters of Prajapati in the Atharvaveda.

Which of the statements given above are correct?

Options: A. 1 and 2 only
B. 2 and 3 only
C. 1 and 3 only
D. 1, 2 and 3

Answer: D

Explanation: - **Core Fact & Solution:** - All three statements are correct: - **Purusha Sukta (Rigveda Mandala X, Hymn 90):** Originates the four Varnas from the cosmic giant *Purusha* (Brahmins from mouth, Kshatriyas from arms, Vaishyas from thighs, Shudras from feet). - **Gayatri Mantra (Rigveda Mandala III, Hymn 62):** Composed by sage **Vishvamitra** in honor of the solar deity **Savitr**. - **Political Assemblies:** Rigveda mentions *Sabha* (council of elders/nobles) and *Samiti* (general assembly of the entire folk). Atharvaveda terms them the **two twin daughters of Prajapati**. - **आयोग का Trap / Ratta Point:** In the early Rigvedic period, women attended *Sabha* and *Vidatha* meetings; but in the Later Vedic period, women were excluded from attending *Sabha*!

Question 322

Subject: General Science (Biology) | **Topic:** Chemical Names of Vitamins

Match List-I (Vitamin) with List-II (Chemical Name) and select the correct answer:

List-I (Vitamin) A. Vitamin A
B. Vitamin B2

C. Vitamin C
D. Vitamin E

List-II (Chemical Name) 1. Riboflavin
2. Retinol
3. Tocopherol
4. Ascorbic Acid

Options: A. A-2, B-1, C-4, D-3
B. A-1, B-2, C-4, D-3
C. A-2, B-4, C-1, D-3
D. A-3, B-1, C-4, D-2

Answer: A

Explanation: - **Core Fact & Solution:** - **Vitamin A: Retinol** (deficiency causes Xerophthalmia and Nyctalopia/Night Blindness). - **Vitamin B2: Riboflavin** (deficiency causes Cheilosis - cracking of lip corners, and Glossitis - red swollen tongue). - **Vitamin C: Ascorbic Acid** (heat-labile, antioxidant; deficiency causes Scurvy). - **Vitamin E: Tocopherol** ("Beauty Vitamin" / anti-sterility vitamin; powerful fat-soluble antioxidant). - **Vitamin K: Phylloquinone / Naphthoquinone** (blood clotting factor). - **Vitamin B6: Pyridoxine** (deficiency causes peripheral neuropathy and convulsions). - **आयोग का Trap / Ratta Point:** Vitamin B2 deficiency causes **Cheilosis** (cracks at mouth corners); Vitamin B3 (Niacin) deficiency causes **Pellagra**!

Question 323

Subject: Uttar Pradesh Special | **Topic:** India State of Forest Report (ISFR) for Uttar Pradesh

Consider the following statements based on the India State of Forest Report (ISFR): 1. The total forest cover and tree cover of Uttar Pradesh constitutes approximately 9.23% of its total geographical area. 2. Sonbhadra district has the largest forest cover in terms of area as well as percentage (~35.24%) in Uttar Pradesh. 3. Bhadohi (Sant Ravidas Nagar) district has the smallest forest cover in terms of area and percentage (~0.37%) in Uttar Pradesh.

Which of the statements given above are correct?

Options: A. 1 and 2 only
B. 2 and 3 only
C. 1 and 3 only
D. 1, 2 and 3

Answer: D

Explanation: - **Core Fact & Solution:** - All three statements are correct: - **Total Green Cover in UP:** Forest cover is **6.15% (14,818 sq km)** + Tree cover is **3.08% (7,421 sq km)** = **Total 9.23% (22,239 sq km)**. - **District Superlatives:** - **Maximum Forest Area: Sonbhadra (2,536 sq km)**, followed by Kheri (1,272 sq km) and Mirzapur (803 sq km). - **Maximum Forest Percentage: Sonbhadra (35.24%)**, followed by Chandauli (21.78%) and Chitrakoot (19.64%). - **Minimum Forest Area & %: Bhadohi (3.71 sq km / 0.37%)**, followed by Mainpuri and Deoria. - **आयोग का Trap / Ratta Point:** Sonbhadra is #1 in BOTH absolute forest area and percentage forest cover! Bhadohi is at the bottom in BOTH!

Question 324

Subject: Environment & Ecology | **Topic:** In-Situ vs Ex-Situ Conservation

Which of the following is an example of **EX-SITU** (off-site) biodiversity conservation? 1. Botanical Gardens 2. Zoological Parks (Zoos) 3. Wildlife Sanctuaries 4. Cryopreservation / Seed Banks

Select the correct answer from the codes given below:

Options: A. 1, 2 and 4 only
B. 1, 2 and 3 only
C. 2, 3 and 4 only
D. 1, 2, 3 and 4

Answer: A

Explanation: - **Core Fact & Solution:** - **In-Situ Conservation (On-Site):** Conserving species in their natural habitats: - National Parks - Wildlife Sanctuaries - Biosphere Reserves - Sacred Groves (*Sarnas* of Jharkhand, *Kavu* of Kerala, *Deorais* of Maharashtra) - Marine Protected Areas (MPAs). - **Ex-Situ Conservation (Off-Site):** Conserving species outside their natural habitats under human management: - **Botanical Gardens** - **Zoological Parks / Aquariums** - **Seed Banks / Gene Banks** - **Cryopreservation** (storing gametes

in liquid nitrogen at $-196^\circ ext{C}$). - **आयोग का Trap / Ratta Point:** Sacred Groves are IN-SITU conservation, NOT ex-situ! Botanical Gardens and Seed Banks are EX-SITU.

Question 325

Subject: Indian Economy | **Topic:** Reserve Bank Open Market Operations (OMOs)

Given below are two statements, one is labelled as Assertion (A) and the other as Reason (R):

Assertion (A): When the Reserve Bank of India wants to absorb excess liquidity from the financial system to curb inflation, it sells government securities in the open market.
Reason (R): Selling government securities transfers cash liquidity from commercial banks into the RBI's accounts, thereby shrinking the credit creation capacity of commercial banks.

Select the correct answer from the codes given below:

Options: A. Both (A) and (R) are true and (R) is the correct explanation of (A)
B. Both (A) and (R) are true, but (R) is not the correct explanation of (A)
C. (A) is true, but (R) is false
D. (A) is false, but (R) is true

Answer: A

Explanation: - **Core Fact & Solution:** - Open Market Operations (OMOs) are outright purchases and sales of Government Securities (G-Secs) by the RBI under Section 17(8) of the RBI Act 1934: - **To Absorb Liquidity (Fight Inflation):** RBI **SELLS** government securities $ ightarrow$ Commercial banks buy them and pay money to RBI $ ightarrow$ Cash reserves of commercial banks contract $ ightarrow$ Money supply falls $ ightarrow$ Interest rates rise. - **To Inject Liquidity (Fight Recession):** RBI **BUYS** government securities $ ightarrow$ RBI injects cash into banking system $ ightarrow$ Banks' lending capacity expands. - **आयोग का Trap / Ratta Point:** Sale of G-Secs by RBI = CONTRACTS liquidity. Purchase of G-Secs by RBI = EXPANDS liquidity!

Question 326

Subject: Current Affairs | **Topic:** Global Innovation Index (GII) 2024

Consider the following statements regarding the Global Innovation Index (GII) 2024 released by the World Intellectual Property Organization (WIPO): 1. Switzerland was ranked as the most innovative economy in the world for the 14th consecutive year. 2. Sweden and the United States were ranked 2nd and 3rd respectively. 3. India was ranked 39th among 133 economies, marking its highest-ever position on the index.

Which of the statements given above are correct?

Options: A. 1 and 2 only
B. 2 and 3 only
C. 1 and 3 only
D. 1, 2 and 3

Answer: D

Explanation: - **Core Fact & Solution:** - All three statements are correct: - Published by **WIPO (Geneva)** in September 2024; evaluates countries across 7 pillars (Institutions, Human capital & research, Infrastructure, Market sophistication, Business sophistication, Knowledge & tech outputs, Creative outputs). - **Top Rank: Switzerland** (1st), Sweden (2nd), USA (3rd), Singapore (4th), UK (5th). - **India:** Climbed to **39th rank** (up from 40th in 2023 and 81st in 2015). India also ranked 1st among Central and Southern Asian economies and 1st among lower-middle-income economies. - **आयोग का Trap / Ratta Point:** India has been an "Innovation Overperformer" for 14 consecutive years relative to its GDP per capita!

Question 327

Subject: Medieval Indian History | **Topic:** Mansabdari System Evolution under Jahangir

Consider the following statements regarding the Mughal Mansabdari system: 1. Akbar established the Mansabdari system, where every officer held a dual rank: *Zat* (personal rank determining status and salary) and *Sawar* (military rank determining cavalry contingent). 2. The *Du-Aspa Sih-Aspa* system was introduced by Emperor Jahangir. 3. Under the *Du-Aspa Sih-Aspa* system, a noble's military contingent was effectively doubled without increasing his *Zat* rank.

Which of the statements given above are correct?

Options: A. 1 and 2 only
B. 2 and 3 only
C. 1 and 3 only
D. 1, 2 and 3

Answer: D

Explanation: - **Core Fact & Solution:** - All three statements are correct: - **Akbar's Framework (1571):** Combined civil and military administration into 33 numerical grades (ranging from Mansab of 10 to 10,000). By 1595–96, ranks were bifurcated into **Zat** (personal status/pay) and **Sawar** (cavalry required to maintain). The *Sawar* rank could never exceed the *Zat* rank! - **Jahangir's Innovation (*Du-Aspa Sih-Aspa*):** Literally "two-horse, three-horse"; allowed trusted nobles to maintain a larger sawar contingent with additional allowance without raising their *Zat* status. - **Shah Jahan's Innovation:** Introduced the *Month-Scale (Mahana)* system (paying salaries for 10, 8, 6, or 4 months instead of full year) due to fiscal shortfalls. - **आयोग का Trap / Ratta Point:** Mansabdari was introduced by **Akbar**; *Du-Aspa Sih-Aspa* was introduced by **Jahangir**; Month-scale was introduced by **Shah Jahan**!

Question 328

Subject: World Geography | **Topic:** Submarine Mid-Oceanic Ridges

Match List-I (Mid-Ocean Ridge / Plateau) with List-II (Ocean Location) and select the correct answer:

List-I (Ridge) A. Mid-Atlantic Ridge
B. Ninety East Ridge
C. Carlsberg Ridge
D. East Pacific Rise

List-II (Ocean Basin) 1. Arabian Sea / Northwest Indian Ocean
2. S-shaped divergent boundary running longitudinally through the Atlantic Ocean
3. Fast-spreading divergent plate boundary across the southeastern Pacific Ocean
4. Remarkable 5,000 km linear north-south volcanic ridge along the 90° E meridian in the Indian Ocean

Options: A. A-2, B-4, C-1, D-3
B. A-1, B-4, C-2, D-3
C. A-2, B-1, C-4, D-3
D. A-3, B-4, C-1, D-2

Answer: A

Explanation: - **Core Fact & Solution:** - **Mid-Atlantic Ridge:** World's longest continuous undersea mountain range (~16,000 km); S-shaped; separates Eurasian/North American and African/South American tectonic plates; surfaces above sea level in Iceland. - **Ninety East Ridge:** Linear submarine ridge along 90° E meridian in the **eastern Indian Ocean**; formed by the Kerguelen hotspot as the Indian plate moved northward. - **Carlsberg Ridge:** Divergent boundary separating the African and Indo-Australian plates in the **Arabian Sea / Central Indian Ocean**. - **East Pacific Rise:** Broad, gently sloping ocean ridge on the Pacific plate boundary with the fastest spreading rates (~15 cm/year). - **आयोग का Trap / Ratta Point:** Ninety East Ridge is located in the **INDIAN OCEAN**, running parallel to the Andaman & Nicobar island arc!

Question 329

Subject: Indian Polity | **Topic:** Election of the Vice-President of India

Consider the following statements regarding the election of the Vice-President of India: 1. The Vice-President is elected by an Electoral College consisting of members of both Houses of Parliament (Article 66). 2. Both elected and nominated members of the Rajya Sabha and Lok Sabha participate in the Vice-Presidential election. 3. Members of State Legislative Assemblies do NOT participate in the election of the Vice-President.

Which of the statements given above are correct?

Options: A. 1 and 2 only
B. 2 and 3 only
C. 1 and 3 only
D. 1, 2 and 3

Answer: D

Explanation: - **Core Fact & Solution:** - All three statements are correct: - **Article 66(1):** Vice-President is elected by an electoral college consisting of the **members of both Houses of Parliament** in accordance with the system of proportional representation by means of the single transferable vote. - **Contrast with Presidential Election:** - *Nominated MPs:* Participate in VP election; but do NOT

participate in President's election. - *State Assemblies (MLAs):* Do NOT participate in VP election; but elected MLAs participate in President's election. - Qualifications: Citizen of India, at least 35 years of age, qualified for election as a member of the Rajya Sabha. - **आयोग का Trap / Ratta Point:** A resolution for the removal of the Vice-President can be introduced **ONLY in the Rajya Sabha** (Article 67(b)), requiring an effective majority in RS and agreed to by Lok Sabha!

Question 330

Subject: General Science (Physics) | **Topic:** Wave Optics & Interference of Light

Consider the following optical phenomena: 1. The brilliant multi-colored patterns observed on thin soap bubbles illuminated by sunlight. 2. The iridescent colors seen on thin oil slicks floating on wet asphalt roads. 3. The alternating bright and dark fringe bands produced in Thomas Young's Double-Slit Experiment.

All three of the above phenomena are direct consequences of which optical principle?

Options: A. Dispersion of light
B. Interference of light waves
C. Total Internal Reflection
D. Polarization of light

Answer: B

Explanation: - **Core Fact & Solution:** - **Interference of Light:** The redistribution of wave energy caused by the superposition of two or more coherent light waves. - **Thin Film Interference (Soap bubbles and Oil films):** When white light strikes a thin film of soap or oil, reflections occur from both the top and bottom surfaces of the film. These reflected waves interfere with each other constructively for certain wavelengths and destructively for others depending on film thickness and angle of incidence, producing brilliant rainbow colors. - **Young's Double Slit Experiment (1801):** Conclusively established the wave nature of light by generating constructive (bright) and destructive (dark) interference fringes on a screen ($y = \frac{n\lambda D}{d}$). - **आयोग का Trap / Ratta Point:** A rainbow in the sky is formed by **Dispersion, Refraction, and Internal Reflection** in water droplets; but colors on a soap bubble or oil film are caused by **INTERFERENCE**!

Question 331

Subject: Uttar Pradesh Special | **Topic:** Minerals of Uttar Pradesh & Famous Belts

Match List-I (Mineral in UP) with List-II (Specific Mining Belt / District) and select the correct answer:

List-I (Mineral) A. Silica Sand (Glass Sand)
B. Limestone
C. Diaspore & Pyrophyllite
D. Copper

List-II (District / Belt) 1. Sonrai area in Lalitpur district
2. Kajrahat and Guruma-Kanach in Sonbhadra district
3. Shankargarh and Lohgara in Prayagraj district
4. Jhansi, Lalitpur, Hamirpur, and Mahoba districts

Options: A. A-3, B-2, C-4, D-1
B. A-2, B-3, C-4, D-1
C. A-3, B-1, C-4, D-2
D. A-4, B-2, C-3, D-1

Answer: A

Explanation: - **Core Fact & Solution:** - **Glass Sand (Silica Sand):** UP ranks high in glass sand, mined in **Shankargarh and Lohgara** (Prayagraj) and Yamuna/Ganga riverbeds. - **Limestone:** Mined in **Sonbhadra** (Kajrahat, Guruma-Kanach-Basahari) and Mirzapur, supplying raw materials to Churk and Dalla cement plants. - **Diaspore & Pyrophyllite:** UP accounts for a major share of India's diaspore and pyrophyllite, mined across **Jhansi, Lalitpur, Hamirpur, and Mahoba** (used in high-temperature refractories). - **Copper:** Mined in the **Sonrai area of Lalitpur** district. - **आयोग का Trap / Ratta Point:** Andalusite and Calcite are mined in **Sonbhadra**; China clay is found in **Sonbhadra and Banda**!

Question 332

Subject: Environment & Ecology | **Topic:** National Clean Air Programme (NCAP)

Consider the following statements regarding the National Clean Air Programme (NCAP): 1. It was launched by the Ministry of Environment, Forest and Climate Change in January 2019 as a national-level strategy to tackle air pollution. 2. It initially targeted a 20% to 30% reduction in particulate matter concentrations (PM_{10} and $PM_{2.5}$) by 2024, using 2017 as the base year. 3. The revised target aims for up to a 40% reduction in particulate matter concentrations by 2026 across 131 non-attainment cities.

Which of the statements given above are correct?

Options: A. 1 and 2 only
B. 2 and 3 only
C. 1 and 3 only
D. 1, 2 and 3

Answer: D

Explanation: - **Core Fact & Solution:** - All three statements are correct: - **Launch:** January 2019; first comprehensive pan-India framework for air quality improvement. - **Non-Attainment Cities:** Cities that consistently failed to meet the National Ambient Air Quality Standards (NAAQS) for 5 consecutive years. Initially 102 cities, later expanded to **131 cities** (including 17 cities in Uttar Pradesh such as Lucknow, Kanpur, Agra, Varanasi, Prayagraj, Ghaziabad, Noida). - **Targets:** In September 2022, the Central Government upgraded the target to achieve **up to 40% reduction or meet national standards (60 $\mu g/m^3$ for PM_{10}) by 2025–26**. - **आयोग का Trap / Ratta Point:** PRANA (Portal for Regulation of Air-pollution in Non-Attainment cities) is the official monitoring portal under NCAP!

Question 333

Subject: Modern Indian History | **Topic:** Revolutionary Conspiracy Trials

Match List-I (Conspiracy Trial) with List-II (Year & Associated Revolutionary) and select the correct answer:

List-I (Conspiracy Case) A. Alipore Bomb Case
B. Delhi-Lahore Conspiracy Case
C. Kakori Train Action
D. Lahore Conspiracy Case

List-II (Year & Revolutionary) 1. 1912; Rash Behari Bose & Sachindra Nath Sanyal (bomb thrown at Viceroy Lord Hardinge)
2. 1908; Aurobindo Ghosh & Barindra Kumar Ghosh (defended brilliantly by C.R. Das)
3. 1929–1930; Bhagat Singh, Sukhdev, and Rajguru (Saunders murder & assembly bombing)
4. 1925; Ram Prasad Bismil, Ashfaqullah Khan, Roshan Singh, Rajendra Lahiri

Options: A. A-2, B-1, C-4, D-3
B. A-1, B-2, C-4, D-3
C. A-2, B-4, C-1, D-3
D. A-3, B-1, C-4, D-2

Answer: A

Explanation: - **Core Fact & Solution:** - **Alipore Bomb Case (1908):** Following Khudiram Bose and Prafulla Chaki's bomb attack on Kingsford at Muzaffarpur; Aurobindo Ghosh was defended by Chittaranjan Das and acquitted, while Barindra Ghosh was sentenced to transportation for life to Cellular Jail. - **Delhi Conspiracy Case (1912):** Bomb hurled at Viceroy Lord Hardinge II during royal procession at Chandni Chowk; orchestrated by Rash Behari Bose (who fled to Japan disguised as P.N. Tagore). - **Kakori Train Action (9 August 1925):** HRA revolutionaries looted the No. 8 Down train near Lucknow; 4 revolutionaries hanged in Dec 1927: **Ram Prasad Bismil (Gorakhpur jail), Ashfaqullah Khan (Faizabad jail), Roshan Singh (Naini/Allahabad jail), Rajendra Lahiri (Gonda jail)**. - **Lahore Conspiracy Case (1929):** Bhagat Singh, Rajguru, Sukhdev sentenced to death and hanged on **23 March 1931**; Jatin Das died on 64th day of hunger strike. - **आयोग का Trap / Ratta Point:** Execution jails of Kakori martyrs: Bismil = **Gorakhpur**; Ashfaqullah = **Faizabad**; Roshan Singh = **Allahabad/Naini**; Rajendra Lahiri = **Gonda** (hanged 2 days early on 17 Dec)!

Question 334

Subject: Indian Economy | **Topic:** Public Sector Disinvestment Policy

Which department under the Ministry of Finance is responsible for managing government equity investments and implementing the disinvestment program in Central Public Sector Enterprises (CPSEs)?

Options: A. Department of Economic Affairs (DEA)
B. Department of Investment and Public Asset Management (DIPAM)
C. Department of Public Enterprises (DPE)
D. Department of Financial Services (DFS)

Answer: B

Explanation: - **Core Fact & Solution:** - **DIPAM (Department of Investment and Public Asset Management):** - Originally set up as the Department of Disinvestment in 1999; renamed as **DIPAM** in the Union Budget 2016–17. - Functions under the **Ministry of Finance**. - Mandated with: strategic disinvestment, minority stake sales via Offer for Sale (OFS), IPOs, ETFs (Bharat 22, CPSE ETF), and asset monetization of CPSEs. - Department of Public Enterprises (DPE) was also transferred to the Ministry of Finance in 2021 (previously under Ministry of Heavy Industries). - **आयोग का Trap / Ratta Point:** Strategic Disinvestment involves the sale of a substantial portion of government shareholding (usually 50% or more) along with the transfer of **management control** to a strategic buyer (e.g. sale of Air India to Talace Pvt Ltd / Tata Sons)!

Question 335

Subject: Current Affairs | **Topic:** G20 Summit 2024 & Global Alliance Against Hunger

Consider the following statements regarding the 19th G20 Leaders' Summit held in Rio de Janeiro, Brazil in November 2024: 1. The official motto of the Brazilian G20 Presidency was "Building a Just World and a Sustainable Planet". 2. Brazilian President Luiz Inacio Lula da Silva officially launched the 'Global Alliance Against Hunger and Poverty' at the summit. 3. India joined the Global Alliance Against Hunger and Poverty as a founding member.

Which of the statements given above are correct?

Options: A. 1 and 2 only
B. 2 and 3 only
C. 1 and 3 only
D. 1, 2 and 3

Answer: D

Explanation: - **Core Fact & Solution:** - All three statements are correct: - **19th G20 Summit (18–19 November 2024, Rio de Janeiro, Brazil):** - Motto: "*Building a Just World and a Sustainable Planet*". - Three priorities: Social inclusion and the fight against hunger/poverty; Energy transition and sustainable development; Reform of global governance institutions. - **Global Alliance Against Hunger and Poverty:** Flagship initiative of President Lula da Silva; joined by over 148 members (including 82 countries, AU, EU, international financial organizations, and philanthropic foundations). - India strongly supported and joined the initiative, sharing experiences of its PM Garib Kalyan Anna Yojana and Direct Benefit Transfers. - **आयोग का Trap / Ratta Point:** The G20 Presidency passed to **South Africa** on December 1, 2024 (marking the first time the G20 summit will be hosted in Africa in 2025)!

Question 336

Subject: Ancient Indian History | **Topic:** Gupta Administrative Terminology

Match List-I (Gupta Administrative Officer) with List-II (Department / Responsibility) and select the correct answer:

List-I (Officer) A. Uparika
B. Vishayapati
C. Sandhivigrahika
D. Mahadandanayaka

List-II (Responsibility) 1. Governor of a Province (*Bhukti*)
2. Minister for Foreign Affairs, War, and Peace (first created by Samudragupta)
3. Head of a District (*Vishaya*)
4. Chief Police / Judicial Executive and Military Commander

Options: A. A-1, B-3, C-2, D-4
B. A-3, B-1, C-2, D-4
C. A-1, B-2, C-3, D-4
D. A-2, B-3, C-1, D-4

Answer: A

Explanation: - **Core Fact & Solution:** - **Uparika:** Provincial governor appointed directly by the emperor to govern a *Bhukti* (province). - **Vishayapati:** District magistrate/collector administering a *Vishaya* (district), assisted by the *Adhishthana Adhikarana* (district council comprising Nagarasreshthi, Sarthavaha, Prathamakulika, and Prathamakayastha). - **Sandhivigrahika:** "Minister of peace and war"; instituted under Samudragupta (Harishena held this title). - **Mahadandanayaka:** Chief justice and general of forces. - **आयोग का Trap / Ratta Point:** In Gupta times, territory was divided into: **Empire -> Bhukti (Province) -> Vishaya (District) -> Vithi (Sub-district) -> Gram (Village)**!

Question 337

Subject: Geography of India | **Topic:** West-Flowing Rivers of India

Consider the following statements regarding the west-flowing rivers of Peninsular India: 1. The Sabarmati river originates from the Dhebar Lake in the Aravalli range in Udaipur district of Rajasthan. 2. The Mahi river originates in Madhya Pradesh and cuts across the Tropic of Cancer twice before emptying into the Gulf of Khambhat. 3. The Luni river originates near Ajmer in the Aravallis and terminates in the marshy expanse of the Rann of Kutch without reaching the open sea.

Which of the statements given above are correct?

Options: A. 1 and 2 only
B. 2 and 3 only
C. 1 and 3 only
D. 1, 2 and 3

Answer: D

Explanation: - **Core Fact & Solution:** - All three statements are correct: - **Sabarmati:** Originates in Aravallis (Udaipur); flows through Rajasthan and Gujarat; Gandhinagar and Ahmedabad are situated on its banks; drains into Gulf of Khambhat. - **Mahi:** Originates in Vindhyas (Dhar, MP); flows through MP, Rajasthan, and Gujarat; is the **only Indian river that crosses the Tropic of Cancer ($23.5^\circ ext{ N}$) twice**. - **Luni (Salt River / Lavanavari):** Originates in Pushkar valley near Ajmer; flows through Thar Desert; water is fresh up to Balotra (Barmer) and becomes saline downstream; classic example of **Inland / Ephemeral Drainage**, disappearing into the Great Rann of Kutch. - **आयोग का Trap / Ratta Point:** Ghaggar in the north and Luni in Rajasthan are India's two most prominent rivers of **Inland Drainage**!

Question 338

Subject: Indian Polity | **Topic:** President's Rule (Article 356) & S.R. Bommai Guidelines

Consider the following statements regarding President's Rule under Article 356: 1. A proclamation of President's Rule must be approved by both Houses of Parliament within two months by a simple majority. 2. The proclamation can be extended for a maximum period of three years with parliamentary approval every six months. 3. In the landmark S.R. Bommai case (1994), the Supreme Court ruled that a proclamation under Article 356 is subject to judicial review.

Which of the statements given above are correct?

Options: A. 1 and 2 only
B. 2 and 3 only
C. 1 and 3 only
D. 1, 2 and 3

Answer: D

Explanation: - **Core Fact & Solution:** - All three statements are correct: - **Grounds:** Article 356 (failure of constitutional machinery in state) and Article 365 (failure to comply with union directions). - **Parliamentary Approval:** Within **two months** by a simple majority in each House. - **Duration:** Valid for 6 months at a time, up to a maximum of **3 years** (beyond 1 year requires meeting two conditions under 44th CAA: National Emergency in operation and Election Commission certifying difficulty in holding assembly elections). - **S.R. Bommai (1994):** 9-judge bench laid down that: Article 356 is an exceptional power; state assembly cannot be dissolved until Parliament approves the proclamation; proclamation is subject to judicial review; if struck down, the court can revive the dismissed government and assembly. - **आयोग का Trap / Ratta Point:** National Emergency (Art 352) requires approval within **1 month by special majority**; President's Rule (Art 356) requires approval within **2 months by simple majority**!

Question 339

Subject: General Science (Biology) | **Topic:** Cellular Elements of Human Blood

Match List-I (Blood Cell Type) with List-II (Biological Feature / Lifespan) and select the correct answer:

List-I (Cell Type) A. Erythrocytes (RBCs)
B. Leucocytes (WBCs)
C. Thrombocytes (Platelets)
D. Monocytes

List-II (Feature / Lifespan) 1. Nucleated; short-lived (1–15 days); defend body against pathogens; divided into granulocytes and agranulocytes
2. Biconcave circular discs lacking nucleus in mature state; contain hemoglobin; lifespan ~120 days

3. Largest white blood cells; kidney-shaped nucleus; differentiate into tissue macrophages
4. Non-nucleated cytoplasmic fragments formed from megakaryocytes; essential for hemostasis and clotting

Options: A. A-2, B-1, C-4, D-3
B. A-1, B-2, C-4, D-3
C. A-2, B-4, C-1, D-3
D. A-3, B-1, C-4, D-2

Answer: A

Explanation: - **Core Fact & Solution:** - **RBCs:** Most abundant ($5–5.5 ext{ million/mm}^3$); lack nucleus, mitochondria, and Golgi apparatus in mammals (to maximize space for hemoglobin); destroyed in the **Spleen ("Graveyard of RBCs")** after ~120 days. - **WBCs:** $6,000–8,000/ ext{mm}^3$; combat infection; Neutrophils are the most numerous (~60–65%); Basophils secrete histamine and heparin; Eosinophils combat allergic reactions. - **Platelets:** $1.5–3.5 ext{ lakh/mm}^3$; pinched off from bone marrow megakaryocytes; release platelet factor 3 for coagulation (dengue causes thrombocytopenia). - **Monocytes:** Largest WBCs (6–8%); phagocytic. - **आयोग का Trap / Ratta Point:** Camels and Llamas are the only mammals that have nucleated, oval RBCs! All other mammals have enucleated circular biconcave RBCs.

Question 340

Subject: Uttar Pradesh Special | **Topic:** Administrative Structure of Uttar Pradesh

Consider the following numbers representing the constitutional and administrative structure of Uttar Pradesh: 1. Total number of Districts: 75 2. Total number of Administrative Divisions: 18 (latest is Aligarh) 3. Total number of Municipal Corporations (Nagar Nigam): 17 (latest is Shahjahanpur) 4. Seats in the Uttar Pradesh Legislative Assembly (Vidhan Sabha): 403

Which of the above administrative figures are correct?

Options: A. 1, 2 and 3 only
B. 1, 3 and 4 only
C. 2, 3 and 4 only
D. 1, 2, 3 and 4

Answer: D

Explanation: - **Core Fact & Solution:** - All four administrative statistics are correct: - **75 Districts & 18 Divisions:** 18th division was Aligarh (carved in 2008). - **17 Municipal Corporations:** 17th is Shahjahanpur (notified in 2018). - **Vidhan Sabha:** 403 elected members (the Anglo-Indian nomination was ended by the 104th CAA 2019). - **Vidhan Parishad:** 100 members. - **Parliamentary Representation: 80 Lok Sabha seats** (highest in India) and **31 Rajya Sabha seats** (highest in India). - **आयोग का Trap / Ratta Point:** Lok Sabha = 80; Rajya Sabha = 31; Vidhan Sabha = 403; Vidhan Parishad = 100!

Question 341

Subject: Environment & Ecology | **Topic:** Water Pollution Parameters (BOD vs COD)

Consider the following statements regarding Biochemical Oxygen Demand (BOD) and Chemical Oxygen Demand (COD): 1. BOD measures the amount of dissolved oxygen consumed by aerobic microorganisms to decompose biologically degradable organic matter in water. 2. COD measures the amount of oxygen required for the chemical oxidation of BOTH biodegradable and non-biodegradable organic pollutants using a strong chemical oxidant like potassium dichromate ($K_2Cr_2O_7$). 3. In any polluted water sample, the COD value is always greater than the BOD value ($ ext{COD} > ext{BOD}$).

Which of the statements given above are correct?

Options: A. 1 and 2 only
B. 2 and 3 only
C. 1 and 3 only
D. 1, 2 and 3

Answer: D

Explanation: - **Core Fact & Solution:** - All three statements are correct: - **BOD (BOD_5 at 20°C):** Measures biological decomposition over 5 days; reflects organic sewage pollution; cannot oxidize toxic or biologically inert compounds (cellulose, lignin, detergents). - **COD:** Determined chemically within 2–3 hours using acidified potassium dichromate ($K_2Cr_2O_7$) and sulphuric acid; oxidizes virtually all organic matter. - **Mathematical Rule:** Because COD oxidizes everything that BOD oxidizes PLUS recalcitrant non-biodegradable pollutants, **COD is ALWAYS greater than or equal to BOD** ($ ext{COD} \ge ext{BOD}$). - **आयोग का Trap / Ratta Point:** A low BOD/COD ratio indicates that the wastewater contains a high proportion of toxic or non-biodegradable chemicals that cannot be treated by biological methods alone!

Question 342

Subject: Modern Indian History | **Topic:** 19th Century Reformers of Western India

Match List-I (Reform Movement / Leader) with List-II (Organization / Historic Role) and select the correct answer:

List-I (Leader / Movement) A. Gopal Hari Deshmukh
B. Dadoba Pandurang
C. Vishnu Shastri Pandit
D. Behramji M. Malabari

List-II (Organization / Contribution) 1. Paramahansa Mandali (1849, Bombay); secret society promoting monotheism and breaking caste taboos
2. Wrote under the pen-name 'Lokahitawadi' in weekly *Prabhakar*, attacking orthodox traditions
3. Founded the Parsi Seva Sadan (1908) and campaigned tirelessly for the Age of Consent Act, 1891
4. Founded the Widow Remarriage Association (1850s) in Maharashtra

Options: A. A-2, B-1, C-4, D-3
B. A-1, B-2, C-4, D-3
C. A-2, B-4, C-1, D-3
D. A-3, B-1, C-4, D-2

Answer: A

Explanation: - **Core Fact & Solution:** - **Gopal Hari Deshmukh (Lokahitawadi):** Judge and social reformer; advocated modernization and equality; wrote 108 essays (*Shatapatre*) in *Prabhakar*. - **Dadoba Pandurang:** Founded **Paramahansa Mandali** in 1849 along with Durgaram Mehtaji; members ate bread cooked by lower-caste persons at secret meetings. - **Vishnu Shastri Pandit:** Prominent champion of widow remarriage in Maharashtra; founded the Widow Remarriage Association in the 1850s. - **Behramji Malabari:** Parsi humanitarian reformer; his relentless agitation against child marriage forced the British government to pass the **Age of Consent Act of 1891** (forbidding marriage of girls below 12 years). - **आयोग का Trap / Ratta Point:** Bal Gangadhar Tilak opposed the Age of Consent Bill 1891, arguing that a foreign colonial government had no right to interfere in Hindu domestic customs!

Question 343

Subject: Indian Economy | **Topic:** Lorenz Curve and Gini Coefficient

Consider the following statements regarding the measurement of income inequality: 1. The Lorenz Curve is a graphical representation of the cumulative distribution of income across a population. 2. The Gini Coefficient is the mathematical ratio of the area between the Line of Perfect Equality and the Lorenz Curve to the total area under the Line of Perfect Equality. 3. A Gini coefficient of 0 indicates absolute mathematical equality, while a Gini coefficient of 1 indicates absolute inequality.

Which of the statements given above are correct?

Options: A. 1 and 2 only
B. 2 and 3 only
C. 1 and 3 only
D. 1, 2 and 3

Answer: D

Explanation: - **Core Fact & Solution:** - All three statements are correct: - **Lorenz Curve (Max O. Lorenz, 1905):** Plots cumulative percentage of population against cumulative percentage of income received. A 45° diagonal represents the "Line of Perfect Equality". - **Gini Coefficient (Corrado Gini, 1912):** $$G = \frac{ ext{Area between Line of Equality and Lorenz Curve}}{ ext{Total Area under Line of Equality}}$$ - **Range:** $0 \le G \le 1$: - **$G = 0$:** Complete equality (everyone has identical income). - **$G = 1$:** Complete inequality (one individual has all income, others zero). - **आयोग का Trap / Ratta Point:** The closer the Lorenz curve bends away from the 45-degree diagonal, the HIGHER the Gini coefficient and the GREATER the income inequality!

Question 344

Subject: Current Affairs | **Topic:** Paris Olympic Games 2024 (Medals Table)

In the final medals tally of the Paris 2024 Olympic Games, which country finished at the top of the leaderboard?

Options: A. United States of America (USA)
B. People's Republic of China

C. Japan
D. France

Answer: A

Explanation: - **Core Fact & Solution:** - The **United States of America (USA)** finished 1st on the medals table with **40 Gold, 44 Silver, and 42 Bronze (Total: 126 medals)**. - **China** finished 2nd with **40 Gold, 27 Silver, and 24 Bronze (Total: 91 medals)**. - (Both USA and China won 40 Gold medals each, but the USA topped the standings by virtue of winning more silver medals: 44 vs 27). - Japan finished 3rd (20 Gold), Australia 4th (18 Gold), and host nation France 5th (16 Gold). - India finished 71st with 6 medals (1 Silver, 5 Bronze). - **आयोग का Trap / Ratta Point:** USA and China tied with exactly 40 Golds each, but the USA edged ahead on Silver medal count!

Question 345

Subject: Medieval Indian History | **Topic:** Mughal Moti Masjids (Agra vs Delhi)

Match List-I (Mughal Mosque) with List-II (Builder & Location) and select the correct answer:

List-I (Mosque) A. Moti Masjid inside Agra Fort
B. Moti Masjid inside Red Fort, Delhi
C. Jama Masjid, Delhi
D. Badshahi Mosque, Lahore

List-II (Builder & Location) 1. Aurangzeb; built inside the Red Fort for personal prayer
2. Shah Jahan; constructed entirely in chaste white marble inside Agra Fort
3. Aurangzeb; largest Mughal mosque built in red sandstone with marble domes
4. Shah Jahan; majestic congregational mosque near Red Fort, Delhi

Options: A. A-2, B-1, C-4, D-3
B. A-1, B-2, C-4, D-3
C. A-2, B-4, C-1, D-3
D. A-3, B-1, C-4, D-2

Answer: A

Explanation: - **Core Fact & Solution:** - **Moti Masjid (Agra Fort):** Built by **Shah Jahan** (1648–1654); made entirely of immaculate white marble. - **Moti Masjid (Delhi Red Fort):** Built by **Aurangzeb** in 1659–1660 as a private prayer chapel for himself and the imperial harem. - **Jama Masjid (Delhi):** Built by **Shah Jahan** (1650–1656); largest congregational mosque of Mughal India. - **Badshahi Mosque (Lahore):** Built by **Aurangzeb** in 1673; grand architectural climax of the Mughal imperial style. - **आयोग का Trap / Ratta Point:** Classic UPPSC trap: - Moti Masjid in AGRA = **Shah Jahan** - Moti Masjid in DELHI = **Aurangzeb**!

Question 346

Subject: World Geography | **Topic:** Major World Canals

Match List-I (Canal) with List-II (Country & Strategic Function) and select the correct answer:

List-I (Canal) A. Corinth Canal
B. Kiel Canal
C. Welland Canal
D. Panama Canal

List-II (Location & Function) 1. Germany; connects North Sea with Baltic Sea through Jutland peninsula
2. Greece; cuts across Isthmus of Corinth connecting Gulf of Corinth with Saronic Gulf
3. Central America; connects Atlantic and Pacific Oceans using 3 sets of locks
4. Canada; bypasses Niagara Falls to connect Lake Erie with Lake Ontario

Options: A. A-2, B-1, C-4, D-3
B. A-1, B-2, C-4, D-3
C. A-2, B-4, C-1, D-3
D. A-3, B-1, C-4, D-2

Answer: A

Explanation: - **Core Fact & Solution:** - **Corinth Canal (Greece):** Spectacular deep, narrow vertical-walled rock cutting across the Isthmus of Corinth (~6.4 km long). - **Kiel Canal (Germany):** World's busiest artificial waterway, linking Brunsbüttel (North Sea) to Kiel-Holtenau (Baltic Sea). - **Welland Canal (Ontario, Canada):** Vital link in the St. Lawrence Seaway; allows ocean-going freighters to bypass the unnavigable 51-meter drop of **Niagara Falls** between Lake Erie and Lake Ontario. - **Panama Canal:** Cuts across Central

America; opened in 1914. - **आयोग का Trap / Ratta Point:** Lake Erie has a higher water elevation than Lake Ontario; the Welland Canal uses 8 lift locks to safely lower ships over the Niagara Escarpment!

Question 347

Subject: Indian Polity | **Topic:** Joint Sitting of Parliament (Article 108)

Consider the following statements regarding a Joint Sitting of both Houses of Parliament: 1. Under Article 108, the President summons a Joint Sitting to resolve a legislative deadlock between the Lok Sabha and the Rajya Sabha. 2. The Speaker of the Lok Sabha presides over a Joint Sitting, and in his absence, the Deputy Speaker of the Lok Sabha presides. 3. If both Speaker and Deputy Speaker are absent, the Chairman of the Rajya Sabha (Vice-President) presides over the Joint Sitting. 4. A Joint Sitting cannot be summoned for a Money Bill or a Constitution Amendment Bill.

Which of the statements given above are correct?

Options: A. 1, 2 and 4 only
B. 1, 3 and 4 only
C. 2, 3 and 4 only
D. 1, 2, 3 and 4

Answer: A

Explanation: - **Core Fact & Solution:** - Statements 1, 2, and 4 are correct: Article 108 summons joint sitting if a bill passed by one house is rejected, unamended for 6 months, or houses disagree on amendments. Bill is passed by simple majority of total members present and voting. - Statement 3 is incorrect: **The Chairman of the Rajya Sabha (Vice-President of India) NEVER presides over a Joint Sitting!** This is because the Vice-President is not a member of either House. In the absence of both Speaker and Deputy Speaker, the **Deputy Chairman of the Rajya Sabha** presides! - **आयोग का Trap / Ratta Point:** Joint sittings have occurred only 3 times in India's history: 1. Dowry Prohibition Bill (1961) 2. Banking Service Commission (Repeal) Bill (1978) 3. Prevention of Terrorism Bill / POTA (2002).

Question 348

Subject: General Science (Chemistry) | **Topic:** Modern Periodic Table Principles

Consider the following statements regarding the Modern Periodic Table: 1. Henry Moseley demonstrated in 1913 that the atomic number (Z) of an element is a more fundamental property than its atomic mass (A). 2. The Modern Periodic Law states that the physical and chemical properties of elements are periodic functions of their atomic numbers. 3. In the modern periodic table, elements are arranged in 18 vertical columns called 'Groups' and 7 horizontal rows called 'Periods'.

Which of the statements given above are correct?

Options: A. 1 and 2 only
B. 2 and 3 only
C. 1 and 3 only
D. 1, 2 and 3

Answer: D

Explanation: - **Core Fact & Solution:** - All three statements are correct: - **Moseley's Law ($\sqrt{ u} = a(Z - b)$):** Proved X-ray spectral frequencies correlate directly with atomic number (nuclear charge Z). - **Mendeleev vs Moseley:** Dmitri Mendeleev (1869) arranged elements by **atomic mass**; Moseley modified it to **atomic number**. - **Architecture:** 7 Periods (Period 1 has 2 elements; Periods 2 & 3 have 8; Periods 4 & 5 have 18; Period 6 has 32; Period 7 is incomplete). 18 Groups (Group 1: Alkali metals; Group 2: Alkaline earths; Group 17: Halogens; Group 18: Noble gases). - **आयोग का Trap / Ratta Point:** Across a period (left to right), atomic radius **DECREASES** and electronegativity **INCREASES**! Down a group, atomic radius **INCREASES**.

Question 349

Subject: Uttar Pradesh Special | **Topic:** One District One Product (ODOP) Master Match

Match List-I (District of UP) with List-II (Designated ODOP Product) and select the correct answer:

List-I (District) A. Gorakhpur
B. Saharanpur
C. Agra
D. Muzaffarnagar

List-II (ODOP Product) 1. Leather Products and Petha
2. Terracotta craft and figurines
3. Jaggery (Gur) manufacturing
4. Wood carving and wooden handicrafts

Options: A. A-2, B-4, C-1, D-3
B. A-4, B-2, C-1, D-3
C. A-2, B-1, C-4, D-3
D. A-3, B-4, C-1, D-2

Answer: A

Explanation: - **Core Fact & Solution:** - **Gorakhpur:** Terracotta pottery, decorative elephants, horse figurines (GI tagged, centered at Bhathat). - **Saharanpur:** Famous worldwide for intricate Sheesham wood carving and furniture. - **Agra:** Leather shoes, footwear, and the confection Petha. - **Muzaffarnagar:** Asia's largest jaggery (*Gur*) market. - **आयोग का Trap / Ratta Point:** Bulandshahr (Khurja) = Ceramics; Mainpuri = Tarkashi craft; Bareilly = Zari-zardozi; Sitapur = Durrie!

Question 350

Subject: Environment & Ecology | **Topic:** IUCN Red List Conservation Categories

Arrange the following IUCN Red List categories in ascending order of extinction risk (from lowest risk to highest risk of extinction): 1. Vulnerable (VU) 2. Critically Endangered (CR) 3. Near Threatened (NT) 4. Endangered (EN)

Select the correct answer from the codes given below:

Options: A. 3 — 1 — 4 — 2
B. 3 — 4 — 1 — 2
C. 1 — 3 — 4 — 2
D. 3 — 1 — 2 — 4

Answer: A

Explanation: - **Core Fact & Solution:** - Ascending hierarchy of threat categories established by IUCN: - Least Concern (LC) - **Near Threatened (NT)** - *Threatened Categories (3):* 1. **Vulnerable (VU):** High risk of extinction in the wild (e.g. Greater One-Horned Rhino, Snow Leopard). 2. **Endangered (EN):** Very high risk of extinction in the wild (e.g. Asiatic Lion, Bengal Tiger, Red Panda). 3. **Critically Endangered (CR):** Extremely high risk of extinction in the wild (e.g. Great Indian Bustard, Gharial, Hangul, Pygmy Hog). - Extinct in the Wild (EW) - Extinct (EX). - **आयोग का Trap / Ratta Point:** The three official "Threatened" categories of IUCN are strictly: **Vulnerable $ ightarrow$ Endangered $ ightarrow$ Critically Endangered**!

Question 351

Subject: Indian Economy | **Topic:** Inflationary Dynamics and Economic Terms

Match List-I (Economic Phenomenon) with List-II (Definition / Real-World Manifestation) and select the correct answer:

List-I (Phenomenon) A. Disinflation
B. Deflation
C. Reflation
D. Shrinkflation

List-II (Definition) 1. Persistent fall in the general price level of goods and services (negative inflation rate)
2. Deliberate monetary/fiscal stimulus to revive price levels and output from a recessionary slump
3. Slowing down of the rate of inflation (e.g., inflation falls from 7% to 4%)
4. Practice of reducing product size or quantity while retaining the same retail selling price

Options: A. A-3, B-1, C-2, D-4
B. A-1, B-3, C-2, D-4
C. A-3, B-2, C-1, D-4
D. A-2, B-1, C-4, D-3

Answer: A

Explanation: - **Core Fact & Solution:** - **Disinflation:** Inflation is still positive, but the *rate* of price increase is decelerating (e.g., prices rise at 3% instead of 6%). - **Deflation:** General price level actually drops ($ ext{Inflation} < 0\%$); harmful because it prompts consumers to delay purchases, triggering economic depression. - **Reflation:** Policy designed to expand output and curb deflation through monetary easing or deficit spending. - **Shrinkflation:** Stealth inflation where the pack size or weight is reduced (e.g. chocolate bar

shrunk from 100g to 80g) while the price remains unchanged. - **आयोग का Trap / Ratta Point:** Disinflation $ eq$ Deflation! In Disinflation, prices are STILL RISING, just at a slower speed. In Deflation, prices are FALLING.

Question 352

Subject: Current Affairs | **Topic:** Space Science (ISRO XPoSat Mission)

Consider the following statements regarding ISRO's XPoSat (X-ray Polarimeter Satellite) mission launched on January 1, 2024: 1. It was launched aboard PSLV-C58 from the Satish Dhawan Space Centre, Sriharikota. 2. It is India's first dedicated polarimetry space observatory to study intense celestial X-ray sources like black holes and neutron stars. 3. India became only the second country in the world (after the USA with NASA's IXPE) to operate a dedicated space-based X-ray polarimetry mission.

Which of the statements given above are correct?

Options: A. 1 and 2 only
B. 2 and 3 only
C. 1 and 3 only
D. 1, 2 and 3

Answer: D

Explanation: - **Core Fact & Solution:** - All three statements are correct: - **Launch:** January 1, 2024, into a 650 km low Earth orbit with 6-degree inclination. - **Science Goal:** Measures polarization of X-rays from cosmic sources (pulsars, black hole binaries, supernova remnants). - **Payloads:** 1. **POLIX (Polarimeter Instrument in X-rays):** Developed by Raman Research Institute (RRI), Bengaluru (8–30 keV range). 2. **XSPECT (X-ray Spectroscopy and Timing):** Developed by Space Astronomy Group at URSC (0.8–15 keV). - **Global Status:** Only USA (NASA's Imaging X-ray Polarimetry Explorer - IXPE launched 2021) and India possess such a satellite. - **आयोग का Trap / Ratta Point:** POLIX was developed by Raman Research Institute (RRI), Bengaluru!

Question 353

Subject: Modern Indian History | **Topic:** Milestones of Mahatma Gandhi's Life

Arrange the following events of Mahatma Gandhi's life in chronological order: 1. Founding of the Phoenix Settlement in South Africa 2. Return to India on Pravasi Bharatiya Divas 3. Signing of the Poona Pact 4. Historic Fast at Yerwada Jail against the Communal Award

Select the correct answer from the codes given below:

Options: A. 1 — 2 — 4 — 3
B. 1 — 2 — 3 — 4
C. 2 — 1 — 4 — 3
D. 1 — 4 — 2 — 3

Answer: A

Explanation: - **Core Fact & Solution:** 1. **Phoenix Settlement (Durban, South Africa, 1904):** Founded after reading John Ruskin's *Unto This Last*; dedicated to community living and printing *Indian Opinion*. 2. **Return to India (9 January 1915):** Returned permanently with Kasturba aboard SS Arabia; landing at Apollo Bunder, Bombay; celebrated as **Pravasi Bharatiya Divas**. 3. **Fast unto Death at Yerwada Jail (20 September 1932):** Begun to protest British Prime Minister Ramsay MacDonald's Communal Award granting separate electorates to Depressed Classes. 4. **Poona Pact (24 September 1932):** Signed between Dr. B.R. Ambedkar and Madan Mohan Malaviya (on behalf of caste Hindus); abandoned separate electorates and replaced them with reserved seats in joint electorates (increased reserved provincial seats from 71 to 147). - **आयोग का Trap / Ratta Point:** Fast began on **20 September 1932**; Poona Pact was signed 4 days later on **24 September 1932**!

Question 354

Subject: Ancient Indian History | **Topic:** Ancient Universities of India

Match List-I (Ancient University) with List-II (Location & Founder / Patron) and select the correct answer:

List-I (University) A. Nalanda University
B. Vikramashila University
C. Vallabhi University
D. Odantapuri University

List-II (Location & Patron) 1. Bhagalpur (Bihar); founded by Pala king Dharmapala; destroyed by Bakhtiyar Khalji
2. Badgaon, Rajgir (Bihar); founded by Gupta emperor Kumaragupta I; Mahayana Buddhism

3. Bihar Sharif (Bihar); founded by Pala king Gopala I
4. Saurashtra (Gujarat); patronized by Maitraka rulers; Hinayana Buddhism center

Options: A. A-2, B-1, C-4, D-3
B. A-1, B-2, C-4, D-3
C. A-2, B-4, C-1, D-3
D. A-3, B-1, C-4, D-2

Answer: A

Explanation: - **Core Fact & Solution:** - **Nalanda:** Founded by **Kumaragupta I** in 5th century CE; residential university with 10,000 students and library *Dharmaganja* (Ratnasagara, Ratnodadhi, Ratnaranjaka); Hiuen Tsang and I-Tsing studied here; destroyed by Muhammad Bakhtiyar Khalji in 1193. - **Vikramashila:** Established by **Dharmapala** (late 8th century) on the banks of Ganga; Vajrayana Buddhism center; Atisha Dipankara was a scholar. - **Vallabhi:** Western Indian counterpart of Nalanda; patronized by Maitrakas; studied secular subjects and Hinayana Buddhism. - **Odantapuri:** Established by **Gopala I** (founder of Pala dynasty) in Magadha. - **आयोग का Trap / Ratta Point:** Nalanda was founded by **Kumaragupta I (Gupta)**; Vikramashila was founded by **Dharmapala (Pala)**!

Question 355

Subject: Geography of India | **Topic:** Saline and Alkaline Soils Reclamation

Consider the following statements regarding Saline and Alkaline soils in India: 1. Saline soils contain high amounts of free salts of sodium, potassium, and magnesium, and are locally known in Uttar Pradesh as *Reh*, *Kallar*, *Usar*, or *Thur*. 2. They develop predominantly in arid and semi-arid tracts and in poorly drained, canal-irrigated command areas with high water tables. 3. Alkaline soils can be chemically reclaimed for agriculture by the application of Gypsum ($CaSO_4 \cdot 2H_2O$) and green manuring with *Dhaincha* (*Sesbania aculeata*).

Which of the statements given above are correct?

Options: A. 1 and 2 only
B. 2 and 3 only
C. 1 and 3 only
D. 1, 2 and 3

Answer: D

Explanation: - **Core Fact & Solution:** - All three statements are correct: - **Formation:** Intense capillary action under hot dry climate draws dissolved sodium and magnesium salts to the surface, leaving white saline crusts (*Reh*). - **Geographical Extent:** Extensive in southwestern and central UP (Aligarh, Mainpuri, Etawah, Kanpur, Unnao, Rae Bareli) due to excessive canal flooding without sub-surface drainage. - **Reclamation: Gypsum ($CaSO_4 \cdot 2H_2O$)** supplies calcium ions that displace sodium ions from the clay complex: $$ ext{Clay-Na} + CaSO_4 ightarrow ext{Clay-Ca} + Na_2SO_4 ext{ (leached out)}$$ Cultivation of acid-tolerant salt-absorbing crops (*Dhaincha*, sugarbeet, barley) restores organic humus. - **आयोग का Trap / Ratta Point:** To reclaim ACIDIC soil, add **LIME / Limestone ($CaCO_3$)**; to reclaim ALKALINE / USAR soil, add **GYPSUM ($CaSO_4 \cdot 2H_2O$)**!

Question 356

Subject: Indian Polity | **Topic:** Suspension of Fundamental Rights (Article 358 vs Article 359)

Match List-I (Emergency Article) with List-II (Effect on Fundamental Rights) and select the correct answer:

List-I (Article) A. Article 358
B. Article 359
C. Article 20 & Article 21

List-II (Effect) 1. Automatic suspension of the six freedoms guaranteed under Article 19 immediately upon proclamation of external emergency
2. Empowers the President to suspend the right to move any court for the enforcement of specified Fundamental Rights by presidential order
3. Can NEVER be suspended or deprived even during the operation of a National Emergency

Options: A. A-1, B-2, C-3
B. A-2, B-1, C-3
C. A-1, B-3, C-2
D. A-3, B-1, C-2

Answer: A

Explanation: - **Core Fact & Solution:** - **Article 358:** Deals ONLY with Article 19. Operates automatically without requiring a separate presidential order; applies to the entire country. Under 44th CAA 1978, Article 19 is suspended ONLY when emergency is proclaimed on grounds of **War or External Aggression**, NOT on Armed Rebellion. - **Article 359:** Deals with the suspension of the *right to move courts* for specified Fundamental Rights; does not suspend the rights themselves, only their judicial enforcement; requires a specific presidential order. - **44th Amendment Act 1978 Inviolability:** The President **CANNOT suspend the enforcement of Articles 20 and 21** under Article 359 under any circumstance! - **आयोग का Trap / Ratta Point:** Article 358 operates AUTOMATICALLY (only for Article 19); Article 359 requires a PRESIDENTIAL ORDER (covers other FRs except Arts 20 & 21)!

Question 357

Subject: General Science (Physics) | **Topic:** Atmospheric Refraction & Day Length

Consider the following statements regarding atmospheric refraction: 1. The Sun is visible to an observer on Earth about 2 minutes before the actual sunrise and about 2 minutes after the actual sunset. 2. The apparent flattening (oval shape) of the Sun's disc at sunrise and sunset is caused by atmospheric refraction. 3. Because of atmospheric refraction, the duration of daylight from sunrise to sunset is increased by approximately 4 minutes every day.

Which of the statements given above are correct?

Options: A. 1 and 2 only
B. 2 and 3 only
C. 1 and 3 only
D. 1, 2 and 3

Answer: D

Explanation: - **Core Fact & Solution:** - All three statements are correct: - **Early Sunrise & Delayed Sunset:** As light from the Sun enters from vacuum into progressively denser layers of the atmosphere, it bends towards the normal. When the Sun is slightly below the actual horizon ($~0.5^\circ$), its refracted rays reach the observer's eye, causing the Sun to appear already above the horizon ~2 minutes early. The same occurs at sunset. - **Total Extra Daylight:** $2 ext{ minutes (sunrise)} + 2 ext{ minutes (sunset)} = \mathbf{4 ext{ minutes}}$. - **Oval Shape:** Rays from the lower rim of the Sun's disc bend more than rays from the upper rim due to the density gradient of air, compressing the vertical diameter and giving it an oval appearance. - **आयोग का Trap / Ratta Point:** The blue color of the sky is due to SCATTERING; but early sunrise/delayed sunset is due to **REFRACTION**!

Question 358

Subject: Uttar Pradesh Special | **Topic:** Uttar Pradesh Annual Budget

Consider the following statements regarding the Uttar Pradesh State Budget: 1. The size of the Uttar Pradesh State Budget has crossed ₹7.36 lakh crore, making it the largest budget ever presented by any State Government in India. 2. The fiscal deficit of the state is maintained within the 3.5% of Gross State Domestic Product (GSDP) limit mandated under the FRBM framework. 3. The budget places top sectoral allocation priority on capital expenditure, infrastructure projects (expressways, airports), and agriculture.

Which of the statements given above are correct?

Options: A. 1 and 2 only
B. 2 and 3 only
C. 1 and 3 only
D. 1, 2 and 3

Answer: D

Explanation: - **Core Fact & Solution:** - All three statements are correct: - UP Finance Minister Suresh Khanna presented a historic budget exceeding **₹7.36 lakh crore (₹7,36,437 crore)**, dedicated to the holistic goal of making Uttar Pradesh a **\$1 Trillion Economy**. - Fiscal Deficit was targeted at **3.46% of GSDP**, well within the 3.5% FRBM ceiling. - Allocations prioritized the Ganga Expressway, Jewar Noida International Airport, Ayodhya International Airport, Metro rail networks, and youth entrepreneurship (Mukhyamantri Yuva Swarojgar). - **आयोग का Trap / Ratta Point:** Uttar Pradesh is the SECOND largest economy among Indian states by GSDP share (contributing over 9.2% of India's GDP, following Maharashtra)!

Question 359

Subject: Environment & Ecology | **Topic:** Bird Sanctuaries of Uttar Pradesh

Match List-I (Bird Sanctuary in UP) with List-II (District) and select the correct answer:

List-I (Bird Sanctuary) A. Suraha Taal (Loknayak Jayaprakash Bird Sanctuary)
B. Samaspur Bird Sanctuary
C. Nawabganj (Shaheed Chandra Shekhar Azad) Bird Sanctuary
D. Sandi Bird Sanctuary

List-II (District in UP) 1. Unnao
2. Ballia
3. Hardoi
4. Rae Bareli

Options: A. A-2, B-4, C-1, D-3
B. A-4, B-2, C-1, D-3
C. A-2, B-1, C-4, D-3
D. A-3, B-4, C-1, D-2

Answer: A

Explanation: - **Core Fact & Solution:** - **Suraha Taal:** Large natural oxbow lake located in **Ballia** district; renamed Loknayak Jayaprakash Narayan Bird Sanctuary. - **Samaspur Bird Sanctuary:** Perennial wetland located in Salon tehsil of **Rae Bareli** district (Ramsar site). - **Nawabganj Bird Sanctuary:** Located in **Unnao** district along the Kanpur-Lucknow highway; renamed Shaheed Chandra Shekhar Azad Bird Sanctuary (Ramsar site). - **Sandi Bird Sanctuary:** Located in **Hardoi** district along the Garra river (Ramsar site). - **आयोग का Trap / Ratta Point:** Patna Bird Sanctuary (smallest bird sanctuary in UP) is in **Etah** district, NOT in Patna, Bihar! Parvati Arga Bird Sanctuary is in **Gonda** district!

Question 360

Subject: Indian Economy | **Topic:** Direct Taxes vs Indirect Taxes

Which of the following is classified as a **DIRECT TAX** levied by the Central Government in India? 1. Corporation Tax 2. Securities Transaction Tax (STT) 3. Goods and Services Tax (GST) 4. Customs Duty

Select the correct answer from the codes given below:

Options: A. 1 and 2 only
B. 1 and 3 only
C. 2 and 4 only
D. 1, 2 and 4 only

Answer: A

Explanation: - **Core Fact & Solution:** - **Direct Taxes (Incidence and impact fall on the same person; cannot be shifted):** - Personal Income Tax - **Corporation Tax** (tax on net profits of domestic and foreign companies) - **Securities Transaction Tax (STT)** (levied on purchase/sale of equities) - Equalisation Levy (tax on digital advertising services) - Capital Gains Tax. - **Indirect Taxes (Tax liability can be shifted to the final consumer):** - **Goods and Services Tax (GST)** - **Customs Duty** (levied on import/export of goods) - Central Excise Duty (on petroleum crude, diesel, petrol, ATF, natural gas). - **आयोग का Trap / Ratta Point:** STT (Securities Transaction Tax) is a DIRECT tax administered by the Central Board of Direct Taxes (CBDT)!

Question 361

Subject: Current Affairs | **Topic:** 16th BRICS Summit Kazan Declaration (2024)

Consider the following statements regarding the 16th BRICS Summit held in Kazan, Russia in October 2024: 1. It was the first summit held following the expansion of the BRICS grouping to include new full members (Egypt, Ethiopia, Iran, UAE). 2. The summit concluded with the adoption of the 'Kazan Declaration' titled "Strengthening Multilateralism for Just Global Development and Security". 3. Leaders endorsed the exploration of a cross-border settlement system (BRICS Pay / BRICS Clear) to trade in local currencies.

Which of the statements given above are correct?

Options: A. 1 and 2 only
B. 2 and 3 only
C. 1 and 3 only
D. 1, 2 and 3

Answer: D

Explanation: - **Core Fact & Solution:** - All three statements are correct: - **Historic Expansion:** 16th BRICS Summit (22–24 October 2024) hosted by President Vladimir Putin in Kazan, Russia; first summit with expanded 9-member format. - **Kazan Declaration:** 43-page

declaration emphasizing reform of the UN Security Council, Bretton Woods institutions, and peaceful diplomatic resolution of conflicts. - **De-Dollarization & Local Currencies:** Agreed to strengthen correspondent banking networks, explore independent cross-border settlement depositaries (BRICS Clear), and welcomed creation of a "Partner Country" category. - **आयोग का Trap / Ratta Point:** On the sidelines of this Kazan Summit, Prime Minister Narendra Modi and Chinese President Xi Jinping held their **first formal bilateral meeting in five years** following the disengagement agreement on the Line of Actual Control (LAC)!

Question 362

Subject: Medieval Indian History | **Topic:** Bhakti Movement in Maharashtra (Varkari Tradition)

Match List-I (Maharashtra Bhakti Saint) with List-II (Literary Composition / Salient Feature) and select the correct answer:

List-I (Saint) A. Sant Dnyaneshwar (Jnanadeva)
B. Sant Tukaram
C. Sant Eknath
D. Samarth Ramdas

List-II (Work / Tradition) 1. Composed *Bhavartha Dipika* (Dnyaneshwari), a pioneering Marathi commentary on Bhagavad Gita at age 15
2. Wrote devotional lyrical hymns (*Abhangas*); contemporary of Shivaji and Jahangir
3. Author of *Dasbodh*; spiritual preceptor (*Guru*) of Chhatrapati Shivaji; founded Dharakari panth
4. Wrote *Bhavartha Ramayana* and introduced *Bharud* dramatic folk songs; insisted on Marathi over Sanskrit

Options: A. A-1, B-2, C-4, D-3
B. A-2, B-1, C-4, D-3
C. A-1, B-4, C-2, D-3
D. A-3, B-2, C-1, D-4

Answer: A

Explanation: - **Core Fact & Solution:** - **Dnyaneshwar (1275–1296):** Laid foundation of Maharashtra Bhakti; authored *Dnyaneshwari* (*Bhavartha Dipika*) and *Amrutanubhav*. - **Tukaram (1608–1650):** Greatest poet of the Varkari sect; worshipped Lord Vitthala (Vithoba) of Pandharpur; composed thousands of *Abhangas*. - **Eknath (1533–1599):** Bridged early and late Bhakti; wrote *Bhavartha Ramayana* and *Eknathi Bhagavata*; proclaimed: "*If Sanskrit was made by God, was Prakrit born of thieves?*" - **Samarth Ramdas (1608–1681):** Worshipped Rama and Hanuman; authored *Dasbodh*; represented the activist *Dharakari* school. - **आयोग का Trap / Ratta Point:** Varkari sect worships **Lord Vithoba (Krishna)** at Pandharpur on the banks of the Bhima (Chandrabhaga) river!

Question 363

Subject: World Geography | **Topic:** Major World Straits and Border Passages

Match List-I (Strait) with List-II (Landmasses Separated) and select the correct answer:

List-I (Strait) A. Palk Strait
B. Bering Strait
C. Strait of Dover
D. Bass Strait

List-II (Landmasses Separated) 1. Separates mainland Australia (Victoria) from the island of Tasmania
2. Separates Tamil Nadu (India) from Jaffna Peninsula (Sri Lanka)
3. Separates Great Britain (England) from mainland Europe (France)
4. Separates Asia (Siberia, Russia) from North America (Alaska, USA)

Options: A. A-2, B-4, C-3, D-1
B. A-1, B-4, C-3, D-2
C. A-2, B-3, C-4, D-1
D. A-4, B-2, C-1, D-3

Answer: A

Explanation: - **Core Fact & Solution:** - **Palk Strait:** Connects Bay of Bengal with Palk Bay / Gulf of Mannar; separates **India and Sri Lanka**. - **Bering Strait:** Connects Arctic Ocean (Chukchi Sea) with Pacific Ocean (Bering Sea); separates **Russia (Chukchi Peninsula) from USA (Alaska)**; International Date Line passes through it. - **Strait of Dover:** Narrowest part of the English Channel (34 km); separates **Dover (UK) from Calais (France)**; underlain by Channel Tunnel. - **Bass Strait:** Separates **mainland Australia from the island of Tasmania**. - **आयोग का Trap / Ratta Point:** Torres Strait separates **Australia from Papua New Guinea**!

Question 364

Subject: Indian Polity | **Topic:** Seventh Schedule & 42nd Amendment Transfers

Under the 42nd Constitutional Amendment Act, 1976, how many subjects were transferred from the State List to the Concurrent List?

Options: A. 3
B. 4
C. 5
D. 6

Answer: C

Explanation: - **Core Fact & Solution:** - The 42nd Constitutional Amendment Act 1976 transferred **FIVE subjects** from the State List (List II) to the Concurrent List (List III): 1. **Education** 2. **Forests** 3. **Weights and Measures** 4. **Protection of Wild Animals and Birds** 5. **Administration of Justice; constitution and organization of all courts except the Supreme Court and the High Courts**. - This historic shift empowered Parliament to legislate nationally on school education, higher education, national forest conservation (Forest Conservation Act 1980), and wildlife protection. - **आयोग का Trap / Ratta Point:** Memorize the 5 transferred subjects: **Education, Forests, Weights & Measures, Wildlife Protection, and Administration of Subordinate Courts**!

Question 365

Subject: General Science (Biology) | **Topic:** Human Genetics and Chromosomal Basis of Sex

Consider the following statements regarding human genetics: 1. Human somatic cells contain 23 pairs of chromosomes (46 chromosomes): 22 pairs of autosomes and 1 pair of allosomes (sex chromosomes). 2. Human females are homogametic, producing ova carrying only the X chromosome ($22 + X$). 3. Human males are heterogametic, producing two types of sperms in equal proportions: $22 + X$ and $22 + Y$. 4. Genetically, the sex of the child is determined entirely by the genetic contribution of the father.

Which of the statements given above are correct?

Options: A. 1, 2 and 3 only
B. 2, 3 and 4 only
C. 1, 3 and 4 only
D. 1, 2, 3 and 4

Answer: D

Explanation: - **Core Fact & Solution:** - All four statements are correct: - **Chromosomal Constitution:** Females = $44 + XX$; Males = $44 + XY$. - **Sex Determination:** Since the mother can only donate an X chromosome, the genetic sex is determined at fertilization by which sperm fertilizes the ovum: - If an X-bearing sperm fertilizes $ ightarrow 44 + XX ightarrow$ **Female child**. - If a Y-bearing sperm fertilizes $ ightarrow 44 + XY ightarrow$ **Male child**. - The biological father is 100% responsible for the genetic sex of the offspring (debunking social taboos blaming women for female children). - **आयोग का Trap / Ratta Point:** Turner's Syndrome is monosomy ($44 + X0$, 45 chromosomes, sterile female); Klinefelter's Syndrome is trisomy ($44 + XXY$, 47 chromosomes, sterile male with gynecomastia); Down's Syndrome is trisomy of **Chromosome 21**!

Question 366

Subject: Uttar Pradesh Special | **Topic:** Historical Fairs & Commemorative Shrines

Match List-I (Historic Fair of UP) with List-II (District & Origin) and select the correct answer:

List-I (Fair) A. Shakumbhari Devi Fair
B. Dadri Cattle Fair
C. Syed Salar Masud Ghazi Fair
D. Nauchandi Fair

List-II (District & Origin) 1. Ballia; ancient fair named after sage Bhrigu's disciple Dardar
2. Saharanpur; biannual Navratri fair at the foot of Shivaliks
3. Meerut; historic month-long post-Holi fair celebrating communal harmony
4. Bahraich; celebrated at the historic 11th-century Dargah shrine

Options: A. A-2, B-1, C-4, D-3
B. A-1, B-2, C-4, D-3
C. A-2, B-4, C-1, D-3
D. A-3, B-1, C-4, D-2

Answer: A

Explanation: - **Core Fact & Solution:** - **Shakumbhari Devi (Saharanpur):** Located in the Shivalik foothills near Behat; major Shakti pilgrimage where goddess Shakumbhari is worshipped (goddess who fed famine-stricken humanity with vegetarian vegetation). - **Dadri Fair (Ballia):** Second largest cattle fair in India (after Sonpur, Bihar); established in antiquity by sage Bhrigu. - **Syed Salar Masud Ghazi (Bahraich):** Annual Jeth mela held at the Dargah of Ghazi Miyan (nephew of Mahmud of Ghazni), attended by both Hindu and Muslim devotees. - **Nauchandi (Meerut):** Symbol of composite Ganga-Jamuni Tehzeeb; visitors pay homage at both the Nauchandi Devi temple and Bale Miyan shrine. - **आयोग का Trap / Ratta Point:** Magh Mela is at **Prayagraj**; Bateshwar is at **Agra**; Dadri is at **Ballia**; Deva Sharif is at **Barabanki**!

Question 367

Subject: Environment & Ecology | **Topic:** Ecological Food Chains & Energy Pathways

Given below are two statements, one is labelled as Assertion (A) and the other as Reason (R):

Assertion (A): In a terrestrial forest ecosystem, a substantially larger fraction of energy flows through the Detritus Food Chain (DFC) than through the Grazing Food Chain (GFC).
Reason (R): A vast proportion of primary plant biomass (leaves, branches, roots, trunks) is not consumed by living herbivores, but falls to the ground as litter and dead organic matter to be decomposed by bacteria and fungi.

Select the correct answer from the codes given below:

Options: A. Both (A) and (R) are true and (R) is the correct explanation of (A)
B. Both (A) and (R) are true, but (R) is not the correct explanation of (A)
C. (A) is true, but (R) is false
D. (A) is false, but (R) is true

Answer: A

Explanation: - **Core Fact & Solution:** - Assertion is true: In forests, ~90% of primary production enters the detritus pathway. - Reason is true: Living herbivores (deer, insects) consume only about 10% of green standing biomass. Massive wood, bark, roots, and leaf litter fall into the soil horizon (*duff/humus*), where saprophytic fungi and bacteria break down complex lignocellulose, releasing nutrients and driving the Detritus Food Chain. - In deep aquatic ecosystems, by contrast, microscopic phytoplankton are grazed almost immediately by zooplankton, making the Grazing Food Chain dominant. - **आयोग का Trap / Ratta Point:** Aquatic ecosystem = **GFC is dominant**; Terrestrial forest ecosystem = **DFC is dominant**!

Question 368

Subject: Modern Indian History | **Topic:** Charter Acts and End of EIC Commercial Monopoly

Match List-I (Charter Act) with List-II (Core Commercial & Administrative Provision) and select the correct answer:

List-I (Charter Act) A. Charter Act of 1813
B. Charter Act of 1833
C. Charter Act of 1853
D. Regulating Act of 1773

List-II (Provision) 1. Completely abolished the commercial monopoly of East India Company (including tea and China trade); Governor-General of Bengal became Governor-General of India
2. Ended Company's commercial trade monopoly EXCEPT in Tea trade and trade with China; allocated ₹1 lakh for education
3. Separated legislative and executive functions of Governor-General's Council; introduced open competition for civil services
4. First written constitution for British rule in India; established Supreme Court at Calcutta; Governor of Bengal made Governor-General

Options: A. A-2, B-1, C-3, D-4
B. A-1, B-2, C-3, D-4
C. A-2, B-3, C-1, D-4
D. A-4, B-1, C-3, D-2

Answer: A

Explanation: - **Core Fact & Solution:** - **Charter Act 1813:** Deprived Company of commercial monopoly in India, opening trade to all British merchants, EXCEPT retained monopoly in **Tea trade and trade with China**; allowed Christian missionaries to enter India; sanctioned ₹1,00,000 annually for education. - **Charter Act 1833: COMPLETELY abolished all commercial activities** of the Company (ended tea and China trade); transformed Company into purely an administrative body; Lord William Bentinck became first Governor-General of India; Macaulay Law Commission appointed. - **Charter Act 1853:** Added 6 legislative councillors creating Indian (Central) Legislative Council; introduced open merit competition for ICS. - **Regulating Act 1773:** Warren Hastings appointed Governor-General;

established Supreme Court at Fort William (1774). - **आयोग का Trap / Ratta Point:** Partial end of trade monopoly = **1813**; Complete termination of all trade monopoly = **1833**!

Question 369

Subject: Indian Economy | **Topic:** Gross Value Added (GVA) at Basic Prices

Consider the following statements regarding the national income accounting methodology adopted in India since 2015: 1. Headline economic growth is measured in terms of Real GDP at constant market prices, rather than GDP at factor cost. 2. The relationship between GDP at market prices and Gross Value Added (GVA) at basic prices is given by: $ ext{GDP}{ ext{Market Prices}} = ext{GVA}{ ext{Basic Prices}} + ext{Product Taxes} - ext{Product Subsidies}$. 3. The base year for national income calculation is currently 2011–12.

Which of the statements given above are correct?

Options: A. 1 and 2 only
B. 2 and 3 only
C. 1 and 3 only
D. 1, 2 and 3

Answer: D

Explanation: - **Core Fact & Solution:** - All three statements are correct: - In January 2015, the Central Statistics Office (CSO, now NSO) aligned Indian national accounting with the international System of National Accounts (SNA 2008): - Shifted headline GDP from Factor Cost to **Market Prices**. - Shifted sectoral production analysis from Factor Cost to **GVA at Basic Prices**. - Equation: $$ ext{GDP at Market Prices} = ext{GVA at Basic Prices} + ext{Net Product Taxes (Taxes - Subsidies)}$$ - Base year updated from 2004–05 to **2011–12**. - **आयोग का Trap / Ratta Point:** GVA captures the SUPPLY side (income earned by factors of production + net production taxes); GDP captures the DEMAND side (final expenditure on goods and services)!

Question 370

Subject: Current Affairs | **Topic:** International Rankings & Human Rights Reports

Match List-I (Global Report / Index) with List-II (Top Performing Country in 2024) and select the correct answer:

List-I (Report / Index 2024) A. World Happiness Report
B. Global Gender Gap Report
C. Global Innovation Index
D. Environmental Performance Index (EPI)

List-II (Top Rank 2024) 1. Iceland
2. Finland
3. Estonia
4. Switzerland

Options: A. A-2, B-1, C-4, D-3
B. A-1, B-2, C-4, D-3
C. A-2, B-4, C-1, D-3
D. A-3, B-1, C-4, D-2

Answer: A

Explanation: - **Core Fact & Solution:** - **World Happiness Report 2024 (UN SDSN): Finland** ranked 1st for the 7th consecutive year. - **Global Gender Gap Report 2024 (WEF): Iceland** ranked 1st for the 15th consecutive year. - **Global Innovation Index 2024 (WIPO): Switzerland** ranked 1st for the 14th consecutive year. - **Environmental Performance Index 2024 (Yale & Columbia Universities): Estonia** ranked 1st globally, followed by Luxembourg and Germany. - **आयोग का Trap / Ratta Point:** Keep these four locked: Happiness = **Finland**; Gender Gap = **Iceland**; Innovation = **Switzerland**; EPI = **Estonia**!

Question 371

Subject: Ancient Indian History | **Topic:** Buddhist Canonical Literature (Tripitakas)

Match List-I (Pitaka of Buddhist Canon) with List-II (Subject Matter / Contents) and select the correct answer:

List-I (Pitaka) A. Sutta Pitaka
B. Vinaya Pitaka
C. Abhidhamma Pitaka

List-II (Subject Matter) 1. Rules of monastic discipline, monastic ethics, and conduct for monks and nuns
2. Discourses and sermons of the Buddha on ethical and doctrinal matters (divided into 5 Nikayas)
3. Profound philosophical, metaphysical, and psychological analysis of the Dhamma (compiled at 3rd Council)

Options: A. A-2, B-1, C-3
B. A-1, B-2, C-3
C. A-2, B-3, C-1
D. A-3, B-1, C-2

Answer: A

Explanation: - **Core Fact & Solution:** - The *Tripitakas* (Three Baskets) written in Pali: - **Sutta Pitaka:** Recited by **Ananda** at the 1st Buddhist Council (Rajgriha, 483 BCE); contains Buddha's religious teachings across 5 Nikayas (Digha, Majjhima, Samyutta, Anguttara, Khuddaka Nikaya - which includes *Dhammapada* and *Jatakas*). - **Vinaya Pitaka:** Recited by **Upali** at the 1st Council; contains 227 rules of monastic discipline (*Patimokkha*), Suttavibhanga, and Khandhaka. - **Abhidhamma Pitaka:** Compiled at the 3rd Buddhist Council (Pataliputra, 250 BCE) under **Moggaliputta Tissa**; contains 7 books (including *Kathavatthu*). - **आयोग का Trap / Ratta Point:** Sutta and Vinaya were compiled at the **1st Council (Rajgriha)**; Abhidhamma was compiled at the **3rd Council (Pataliputra)**!

Question 372

Subject: Geography of India | **Topic:** Mountain Passes of Northern and Northeastern India

Match List-I (Himalayan Pass) with List-II (State / UT & Connectivity) and select the correct answer:

List-I (Pass) A. Banihal Pass
B. Rohtang Pass
C. Jelep La
D. Bomdila

List-II (Location & Connectivity) 1. Himachal Pradesh; Pir Panjal range; connects Kullu Valley with Lahaul and Spiti; site of Atal Tunnel
2. Jammu & Kashmir; Pir Panjal range; connects Jammu with Srinagar; site of Jawahar Tunnel
3. Arunachal Pradesh; Great Himalayas; connects Tezpur/Guwahati with Tawang and Tibet
4. Sikkim; Chumbi Valley; connects Kalimpong/Gangtok with Lhasa (Tibet)

Options: A. A-2, B-1, C-4, D-3
B. A-1, B-2, C-4, D-3
C. A-2, B-4, C-1, D-3
D. A-3, B-1, C-4, D-2

Answer: A

Explanation: - **Core Fact & Solution:** - **Banihal Pass (J&K):** Located on Pir Panjal range (2,832 m); carries NH-44 through the Jawahar Tunnel connecting Jammu with the Kashmir Valley. - **Rohtang Pass (HP):** Connects Kullu with Lahaul-Spiti; underlain by the 9.02 km long **Atal Tunnel** (world's longest highway tunnel above 10,000 feet). - **Jelep La (Sikkim):** Connects Sikkim with Lhasa via the Chumbi Valley. - **Bomdila (Arunachal Pradesh):** Connects western Arunachal with Lhasa; strategic defense route to Tawang. - **आयोग का Trap / Ratta Point:** Nathu La and Jelep La are in **Sikkim**; Mana, Niti, and Lipulekh are in **Uttarakhand**; Shipki La and Bara Lacha La are in **Himachal Pradesh**!

Question 373

Subject: Indian Polity | **Topic:** Kesavananda Bharati Judgment (1973)

Consider the following statements regarding the landmark Kesavananda Bharati v. State of Kerala judgment: 1. It was decided on April 24, 1973, by the largest bench in the history of the Supreme Court of India, comprising 13 judges. 2. The judgment was delivered by a narrow majority of 7 to 6, headed by Chief Justice S.M. Sikri. 3. The Supreme Court upheld the constitutional validity of the 24th Amendment Act, 1971, ruling that Parliament can amend any part of the Constitution, including Fundamental Rights, provided it does not alter the "Basic Structure".

Which of the statements given above are correct?

Options: A. 1 and 2 only
B. 2 and 3 only
C. 1 and 3 only
D. 1, 2 and 3

Answer: D

Explanation: - **Core Fact & Solution:** - All three statements are correct: - **Date & Bench:** Decided on **24 April 1973** (same day Sachin Tendulkar was born); 13-judge constitutional bench presided by Chief Justice S.M. Sikri. - **Majority:** Narrow 7:6 majority (Sikri, Shelat, Grover, Hegde, Mukherjea, Reddy, and Khanna in majority; Ray, Palekar, Mathew, Beg, Dwivedi, and Chandrachud in minority). Justice H.R. Khanna's opinion tipped the scales. - **Basic Structure Doctrine:** Overruled *Golaknath (1967)*; affirmed Parliament's amending power under Article 368 is broad but not unlimited; Parliament cannot destroy the essential framework or basic features of the Constitution (supremacy of Constitution, republican and democratic form, secularism, separation of powers, federalism, judicial review). - **आयोग का Trap / Ratta Point:** The Supreme Court did NOT define a comprehensive, exhaustive list of what constitutes the "Basic Structure", leaving it to be determined on a case-by-case basis!

Question 374

Subject: General Science (Physics) | **Topic:** Doppler Effect in Sound and Light

Consider the following statements regarding the Doppler Effect: 1. The Doppler Effect is the apparent change in the observed frequency or wavelength of a wave due to relative motion between the source and the observer. 2. When a light-emitting star or galaxy moves away from Earth, its observed spectral lines shift towards longer wavelengths, known as 'Redshift'. 3. The Doppler Effect is utilized in speed radar guns, echocardiography, weather radars, and proving the expansion of the universe.

Which of the statements given above are correct?

Options: A. 1 and 2 only
B. 2 and 3 only
C. 1 and 3 only
D. 1, 2 and 3

Answer: D

Explanation: - **Core Fact & Solution:** - All three statements are correct: - Formulated by Christian Doppler in 1842: - **Sound Waves:** When a train approaches an observer blowing its horn, apparent pitch/frequency is **HIGHER** ($f' > f$). When receding, apparent pitch is **LOWER** ($f' < f$). - **Light Waves:** - Source moving away $ ightarrow$ Wavelength increases $ ightarrow$ **Redshift** (Edwin Hubble used redshift of distant galaxies in 1929 to prove the Universe is expanding). - Source moving closer $ ightarrow$ Wavelength decreases $ ightarrow$ **Blueshift**. - **Applications:** Police traffic RADAR guns, Doppler Weather Radar (cyclone tracking), Medical Doppler Ultrasonography (blood flow measurement). - **आयोग का Trap / Ratta Point:** Redshift means wave is moving AWAY; Blueshift means wave is moving CLOSER!

Question 375

Subject: Uttar Pradesh Special | **Topic:** Industrial Development Authorities of UP

Match List-I (Industrial Development Authority) with List-II (Headquarters / Operational Zone) and select the correct answer:

List-I (Authority) A. NOIDA
B. GIDA
C. SIDA
D. YEIDA

List-II (Zone / City) 1. Gorakhpur Industrial Development Authority
2. New Okhla Industrial Development Authority (Gautam Buddha Nagar)
3. Yamuna Expressway Industrial Development Authority (Jewar Airport zone)
4. Satharia Industrial Development Authority (Jaunpur)

Options: A. A-2, B-1, C-4, D-3
B. A-1, B-2, C-4, D-3
C. A-2, B-4, C-1, D-3
D. A-3, B-1, C-4, D-2

Answer: A

Explanation: - **Core Fact & Solution:** - **NOIDA (New Okhla Industrial Development Authority):** Constituted in 1976 under UP Industrial Area Development Act in Gautam Buddha Nagar. - **GIDA (Gorakhpur Industrial Development Authority):** Industrial hub of eastern UP near Sahjanwa, Gorakhpur. - **SIDA (Satharia Industrial Development Authority):** Located at Satharia in **Jaunpur** district. - **YEIDA (Yamuna Expressway Industrial Development Authority):** Manages industrial, commercial, and residential clusters along the Yamuna Expressway, including the **Noida International Airport at Jewar**. - **UPSIDA (UP State Industrial Development Authority):** Headquartered at **Kanpur**, managing industrial estates across UP. - **आयोग का Trap / Ratta Point:** BIDA (Bundelkhand Industrial Development Authority) was notified in **2023 with headquarters at Jhansi**, modeled after NOIDA to develop Bundelkhand!